

Analysis: Guided Exercises

Instructions

This problem set develops the main ideas of topology, continuity, and multivariate differential calculus that will be used throughout graduate seminars.

Several problems contain guided steps or hints. These are part of the learning process. You should write a complete argument rather than simply answering the individual prompts.

Unless otherwise stated, all norms are Euclidean norms.

1 Topology and Continuity in $\mathbb{R}^n$

Problem 1. Norms and the topology of $\mathbb{R}^n$

For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, define

$$\|x\|_1 = \left(\sum_{i=1}^n |x_i| \right), \quad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

(a) Prove that $\|x\|_1 \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty$.

(b) Deduce that, for a sequence $\{x_k\} \subseteq \mathbb{R}^n$,

$$\|x_k - x\|_2 \rightarrow 0 \iff \|x_k - x\|_\infty \rightarrow 0.$$

(c) Show that the two norms generate the same open sets.

Guidance: For part (c), compare balls. Given $\epsilon > 0$, find $r_1, r_2 > 0$ such that

$$B_{\| \cdot \|_1}(r_1) \subseteq B_{\| \cdot \|_2}(\epsilon), \quad B_{\| \cdot \|_2}(r_2) \subseteq B_{\| \cdot \|_1}(\epsilon).$$

a) for $x = (x_1, \dots, x_n) \in \mathbb{R}^n$

$$\|x\|_\infty = \max_{1 \leq i \leq n} |x_i| = X_m \text{ s.t. } k$$

defined as $X_n = |x_j|$

$$\text{s.t. } |x_j| \geq |x_1|, \dots, |x_j| \geq |x_i|, \dots, |x_j| \geq |x_n|$$

i.e. $i=j$

$$\text{thus } \sum_{i=1}^n |x_i| \geq |x_j| = X_m = \|x\|_\infty$$

$$\text{since } \sum_{i=1}^n |x_i| = \sum_{i=1}^{j-1} |x_i| + |x_j| + \sum_{i=j+1}^n |x_i|$$

s.t.

$$\sum_{i=1}^n |x_i| \geq 0 \text{ by def of abs val}$$

$$\text{thus } \sum_{i=1}^n |x_i| \geq \sum_{i=1}^{j-1} |x_i| + \sum_{i=j+1}^n |x_i| \geq |x_j| \geq 0$$

$$\text{since } \|x\|_2 = \left(\sum_{i=1}^n x_i^2 \right)^{1/2} = (x_1^2 + \dots + x_n^2)^{1/2}$$

$$x_1^2 \geq 0, \dots, x_n^2 \geq 0 \text{ s.t. } x_1^2 + \dots + x_n^2 \geq 0$$

$$\text{+ } x_i^2 = |x_i|^2 \geq 0 \forall x_i$$

$$\text{+ thus } \sum_{i=1}^n x_i^2 = |x_j|^2 + \sum_{i=1}^{j-1} |x_i|^2 + \sum_{i=j+1}^n |x_i|^2 \geq |x_j|^2$$

apply sq root to both sides

$$\left(\sum_{i=1}^n x_i^2 \right)^{1/2} \geq (|x_j|^2)^{1/2}$$

$$(1.4.1) \quad \|x\|_2 \geq |x_j| = \|x\|_\infty$$

to show $\|x\|_2 \leq \sqrt{n} \|x\|_\infty$:

$$|x_i| \leq \|x\|_\infty = \max(x_i) \text{ by def}$$

$$|x_i|^2 = x_i^2 \leq \|x\|_\infty^2$$

summing over n , since true for all x_i

$$n \cdot \|x\|_\infty^2 \geq \|x\|_2^2$$

Summing over n , since true for all x_i :

$$\sum_{i=1}^n x_i^2 \leq \sum_{i=1}^n \|x\|_\infty^2 = n \|x\|_\infty^2$$

since $\|x\|_\infty$ not indexed by i

sq both sides

$$(2a2) \left(\sum_{i=1}^n x_i^2 \right)^{1/2} = \|x\|_2 \leq \sqrt{n} \|x\|_\infty$$

thus, combining 2a2 + 2a2

$$\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty$$

Problem 1. Norms and the topology of $\mathbb{R}^n$

For $x = (x_1, \dots, x_n) \in \mathbb{R}^n$, define

$$\|x\|_2 = \left(\sum_{i=1}^n x_i^2 \right)^{1/2}, \quad \|x\|_\infty = \max_{1 \leq i \leq n} |x_i|.$$

(a) Prove that

$$\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty.$$

(b) Deduce that, for a sequence $(x_k) \subset \mathbb{R}^n$,

$$\|x_k - x\|_2 \rightarrow 0 \iff \|x_k - x\|_\infty \rightarrow 0.$$

for seq. $(x_k) \subset \mathbb{R}^n$

$$\Rightarrow \|x_k - x\|_2 \rightarrow 0 \Rightarrow \|x_k - x\|_\infty \rightarrow 0$$

$$x_k = (x_{k1}, \dots, x_{kn})$$

$$x = (x_1, \dots, x_n)$$

$$\text{define } y_k = (x_{k1} - x_1, \dots, x_{kn} - x_n)$$

$$\text{for a), } \|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty$$

$$\text{so } \|y_k\|_\infty \leq \|y_k\|_2 \leq \sqrt{n} \|y_k\|_\infty$$

$$\text{for c) } 0 \leq \|x\|_\infty \quad \forall x \in \mathbb{R}^n$$

$$\text{thus } 0 \leq \|y_k\|_\infty \leq \|y_k\|_2$$

$$\text{or } 0 \leq \|x_k - x\|_\infty \leq \|x_k - x\|_2$$

$$\text{in limit as } k \rightarrow \infty, \quad \lim_{k \rightarrow \infty} \|x_k - x\|_2 \rightarrow 0$$

$$\Rightarrow 0 \leq \lim_{k \rightarrow \infty} \|x_k - x\|_\infty \leq 0$$

$$\text{thus } \lim_{k \rightarrow \infty} \|x_k - x\|_\infty = 0$$

$$\Leftarrow) \quad \text{if } \|x_k - x\|_\infty \rightarrow 0 \text{ as } k \rightarrow \infty$$

$$\text{for c) } 0 \leq \|x_k - x\|_2 \leq \sqrt{n} \|x_k - x\|_\infty$$

in limit,

$$\lim_{k \rightarrow \infty} \sqrt{n} \|x_k - x\|_\infty = \lim_{k \rightarrow \infty} \|x_k - x\|_2$$

$$\lim_{k \rightarrow \infty} \|x_k - x\|_\infty = 0 \Rightarrow \lim_{k \rightarrow \infty} \sqrt{n} \|x_k - x\|_\infty = 0$$

$$\lim_{k \rightarrow \infty} \|x_k - x\|_\infty \leq \lim_{k \rightarrow \infty} \|x_k - x\|_2 \leq \lim_{k \rightarrow \infty} \sqrt{n} \|x_k - x\|_\infty$$

$$0 \leq \lim_{k \rightarrow \infty} \|x_k - x\|_2 \leq 0$$

$$\Rightarrow \lim_{k \rightarrow \infty} \|x_k - x\|_2 = 0$$

$$0 \leq \lim_{k \rightarrow \infty} \|x_k - x\|_2 \leq 0$$

$$\Rightarrow \lim_{k \rightarrow \infty} \|x_k - x\|_2 = 0$$

Thus, $\|x_k - x\|_2 \rightarrow 0 \Rightarrow \|x_k - x\|_\infty \rightarrow 0$

(c) Show that the two norms generate the same open sets.

Guidance: For part (c), compare balls. Given $r > 0$, find $r_1, r_2 > 0$ such that

$$B_{\infty,r}(x) \subseteq B_2(r) \quad B_{2,r}(x) \subseteq B_{\infty,r}(x)$$

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For $r \in \mathbb{R} : r > 0$,

$$\text{let } B_{\infty,r}(x) = \{y \in \mathbb{R}^n : \|y - x\|_\infty < r\}$$

$$B_{2,r}(x) = \{y \in \mathbb{R}^n : \|y - x\|_2 < r\}$$

$$\text{from d) } \|y - x\|_\infty \leq \|y - x\|_2 \leq \sqrt{2} \|y - x\|_\infty$$

$$\text{to show: } B_{\infty,r}(x) \subseteq B_{2,r}(x)$$

if $y \in B_{\infty,r}(x)$, then $\|y - x\|_\infty \leq r$,

$$\|y - x\|_2 \leq \sqrt{2} \|y - x\|_\infty < \sqrt{2} r$$

pick $r_1 = r/\sqrt{2}$

$$\|y - x\|_2 < \sqrt{2} r_1 = \sqrt{2} \frac{r}{\sqrt{2}} = r$$

thus $y \in B_{2,r}(x)$

$$\text{to show: } B_{2,r_2}(x) \subseteq B_{\infty,r}(x)$$

if $y \in B_{2,r_2}(x)$, then $\|y - x\|_2 < r_2$

$$\|y - x\|_\infty \leq \|y - x\|_2 < r_2$$

pick $r_2 = r$

$$\|y - x\|_\infty < r$$

thus $y \in B_{\infty,r}(x)$

Finding the Radii r_1 and r_2 for Part (c)

Using the norm equivalence from part (d), $\|y - x\|_\infty \leq \|y - x\|_2 \leq \sqrt{n} \|y - x\|_\infty$:

- Inclusion 1: $B_{\infty,r_1}(x) \subseteq B_2(x)$**
 - If $y \in B_{\infty,r_1}(x)$, then $\|y - x\|_\infty < r_1$.
 - By part (d): $\|y - x\|_2 \leq \sqrt{n} \|y - x\|_\infty < \sqrt{n} r_1$.
 - Setting $\sqrt{n} r_1 = r \implies r_1 = \frac{r}{\sqrt{n}}$ ensures $\|y - x\|_2 < r$, so $y \in B_2(x)$.
- Inclusion 2: $B_2(x) \subseteq B_{\infty,r_2}(x)$**
 - If $y \in B_2(x)$, then $\|y - x\|_2 < r_2$.
 - By part (d): $\|y - x\|_\infty \leq \|y - x\|_2 < r_2$.
 - Setting $r_2 = r$ ensures $\|y - x\|_\infty < r$, so $y \in B_{\infty,r}(x)$.

Problem 2. Continuous functions and open or closed sets

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous.

(a) Prove that f is open.

To show: Let $S = \{x \in \mathbb{R}^n : f(x) > c, c \in \mathbb{R}\}$

$\forall x_0 \in S, \exists B_\delta(x_0) \subseteq S$ s.t. $f(B_\delta(x_0)) > c$

for any $x_0 \in S$, take $f(x_0)$

$f(x_0) > c$ by def of S

let $\epsilon = f(x_0) - c$

$f(x_0) > c \Rightarrow f(x_0) - c > 0$

Since f is cont at x_0 , for $\epsilon > 0$

$\exists \delta > 0$ s.t. $\forall x \in B_\delta(x_0)$

$|f(x) - f(x_0)| < \epsilon \Rightarrow |f(x) - f(x_0)| < \epsilon$

$-(f(x_0) - c) < f(x) - f(x_0) < f(x_0) - c$

$f(x) > c$

$\therefore B_\delta(x_0) \subseteq S$

$\therefore f$ is open

$\therefore f$ is open

$\therefore f$ is open

An open set and a closed set in Euclidean space $\mathbb{R}^n$ are formally defined through neighborhoods (open balls) and complementation, respectively.

1. Open Set

- Definition:** A set $U \subseteq \mathbb{R}^n$ is open if for every point $x \in U$, there exists an $\epsilon > 0$ such that the open ball centered at x is contained entirely within U .

- Intuition:** Every point in the set has an "envelope" of surrounding space that also belongs to the set, meaning the set contains none of its boundary points.
- Examples:**
 - Open intervals $(a, b) \subset \mathbb{R}$
 - Open half-spaces $\{x \in \mathbb{R}^n \mid a^T x < b\}$
 - The interior of an open ball $\{x \in \mathbb{R}^n \mid \|x - x_0\| < r\}$

2. Closed Set

- Primary Definition (via Complement):** A set $F \subseteq \mathbb{R}^n$ is closed if its complement $F^c = \mathbb{R}^n \setminus F$ is an open set.
- Equivalent Sequence Characterization:** A set F is closed if and only if it contains all of its limit points. That is, for any sequence $(x_k) \subseteq F$ such that $\lim_{k \rightarrow \infty} x_k = x$, the limit satisfies $x \in F$.
- Intuition:** The set includes all of its boundary points.
- Examples:**
 - Closed intervals $[a, b] \subset \mathbb{R}$
 - Closed half-spaces $\{x \in \mathbb{R}^n \mid a^T x \leq b\}$
 - The closed Euclidean ball $\{x \in \mathbb{R}^n \mid \|x - x_0\| \leq r\}$

Key Topological Distinctions

- Not Mutually Exclusive:** "Open" and "closed" are not opposite binary states. A set can be:
 - Both open and closed ("Clopen"):** $\emptyset$ and the whole space $\mathbb{R}^n$.
 - Neither open nor closed:** Half-open intervals like $[a, b) \subset \mathbb{R}$.
- Continuous Function Characterization:** For a continuous function $f: \mathbb{R}^n \rightarrow \mathbb{R}$:
 - The strict sublevel/superlevel set $\{x \mid f(x) < a\}$ is open.
 - The weak sublevel/superlevel set (such as the upper contour set $U_a = \{x \mid f(x) \geq a\}$) is closed.

From <https://www.google.com/app/264b7558b013b5d0e>

Formal Proof Using the Open Ball Definition

Let $S = \{x \in \mathbb{R}^n : f(x) > c\}$. To show that S is open, we must show that for every $x_0 \in S$, there exists an open ball $B_\delta(x_0) \subseteq S$.

1. Choose an arbitrary point: Let $x_0 \in S$. By definition of the set S , $f(x_0) > c$.

2. Define an ϵ -neighborhood in $\mathbb{R}$: Let $\epsilon = f(x_0) - c$. Since $f(x_0) > c$, we have $\epsilon > 0$.

3. Apply the (ϵ, δ) -definition of continuity: Since $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous at x_0 , for this chosen $\epsilon > 0$, there exists a $\delta > 0$ (such that for all $x \in \mathbb{R}^n$,

$\|x - x_0\| < \delta \implies |f(x) - f(x_0)| < \epsilon$

4. Expand the inequality: The condition $|f(x) - f(x_0)| < \epsilon$ is equivalent to:
 $-\epsilon < f(x) - f(x_0) < \epsilon \implies f(x_0) - \epsilon < f(x) < f(x_0) + \epsilon$

5. Substitute $\epsilon = f(x_0) - c$: Focusing on the lower bound:

$f(x) > f(x_0) - \epsilon = f(x_0) - (f(x_0) - c) = c$

6. Establish the subset inclusion: Thus, for every $x \in B_\delta(x_0) = \{x \in \mathbb{R}^n : \|x - x_0\| < \delta\}$, we have $f(x) > c$, which implies $x \in S$. Therefore, $B_\delta(x_0) \subseteq S$. Since x_0 was arbitrary, S is open. ■

Alternative Topological Approach (Preimage)
 In general topology, this is also shown in one step using the topological definition of continuity (the preimage of every open set is open):

$S = \{x \in \mathbb{R}^n : f(x) > c\} = f^{-1}((c, \infty))$
 Since the interval (c, ∞) is an open set in $\mathbb{R}$ and f is continuous, its preimage $f^{-1}((c, \infty))$ is open in $\mathbb{R}^n$.

from <https://www.google.com/app/264b7558b013b5d0e>

Part (a)

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous and fix $c \in \mathbb{R}$.

Recall the topological characterization of continuity: a function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is continuous if and only if the preimage $f^{-1}(U)$ of every open set $U \subseteq \mathbb{R}$ is open in $\mathbb{R}^n$.

The interval (c, ∞) is an open subset of $\mathbb{R}$. Expressing the set as a preimage gives:

$$\{x \in \mathbb{R}^n : f(x) > c\} = f^{-1}((c, \infty))$$

Because (c, ∞) is open in $\mathbb{R}$ and f is continuous, the preimage $f^{-1}((c, \infty))$ is open in $\mathbb{R}^n$.

$$-(f(x_0) - c) < f(x) - f(x_0) < f(x_0) - c$$

$$L < f(x) < 2f(x_0) - c$$

thus $\forall x \in B_f(x_0)$,

$$B_f(x_0) = \{x \in \mathbb{R}^n : \|x - x_0\| < \delta\}$$

$$f(x) > c,$$

thus S open $\square$

(b) Prove that
is closed.

complement of open set is closed
so if $V = \{x \in \mathbb{R}^n : f(x) \geq c\}$

$$V^c = \{x \in \mathbb{R}^n : f(x) < c\}$$

using same proof as <,
pick $\varepsilon = c - f(x_0)$

$$\forall x \in \mathbb{R}^n : f(x) < c$$

$$f(x) - c < 0$$

by continuity at x_0 , $\forall \varepsilon > 0$

$$\exists \delta > 0 \text{ s.t. } \forall x \in \mathbb{R}^n$$

$$\|x - x_0\| < \delta \Rightarrow |f(x) - f(x_0)| < \varepsilon$$

$$\text{for } x \in V^c, f(x_0) < c$$

$$f(x_0) - c < 0$$

$$\text{multiply by } \begin{matrix} \text{negative} \\ \downarrow \end{matrix} \Rightarrow c - f(x_0) > 0$$

$$\text{so } \varepsilon > 0 \text{ satisfied}$$

$$-\varepsilon < f(x) - f(x_0) < \varepsilon$$

$$-(c - f(x_0)) < f(x) - f(x_0) < c - f(x_0)$$

$$\forall x \in B_f(x_0) = \|x - x_0\| < \delta : f(x) < c$$

thus V^c open

+ V must be closed $\square$

(c) Use these results, rather than returning directly to the definition, to classify the following sets as open or closed:

$$A = \{x \in \mathbb{R}^n : \|x\|^2 < 1\}.$$

$$B = \{x \in \mathbb{R}^n : x^T Q x \leq 1\},$$

where Q is a fixed symmetric $n \times n$ matrix, and

$$C = \{(x, y) \in \mathbb{R}^2 : y = x^2\}.$$

Guidance: Try to express each set as the inverse image of an open or closed subset of $\mathbb{R}$ under a continuous real-valued function.

$$\{x \in \mathbb{R}^n : f(x) > c\} = f^{-1}((c, \infty))$$

Because (c, ∞) is open in $\mathbb{R}$ and f is continuous, the preimage $f^{-1}((c, \infty))$ is open in $\mathbb{R}^n$.

Part (b)

Method 1 (Closed Preimage):

The interval $[c, \infty)$ is closed in $\mathbb{R}$ because its complement in $\mathbb{R}$ is $(-\infty, c)$, which is open. Since the preimage of a closed set under a continuous map is closed:

$$\{x \in \mathbb{R}^n : f(x) \geq c\} = f^{-1}([c, \infty))$$

Thus, the set is closed in $\mathbb{R}^n$.

Method 2 (Complement / Result of Part a):

The complement of $\{x \in \mathbb{R}^n : f(x) \geq c\}$ in $\mathbb{R}^n$ is:

$$\mathbb{R}^n \setminus \{x \in \mathbb{R}^n : f(x) \geq c\} = \{x \in \mathbb{R}^n : f(x) < c\}$$

$$= \{x \in \mathbb{R}^n : -f(x) > -c\}$$

Since f is continuous, $g(x) = -f(x)$ is continuous. By the result in part (a), the set $\{x \in \mathbb{R}^n : g(x) > -c\}$ is open. As its complement is open, $\{x \in \mathbb{R}^n : f(x) \geq c\}$ is closed.

Part (c)

• **Set $A = \{x \in \mathbb{R}^n : \|x\|^2 < 1\}$**

- Define $g: \mathbb{R}^n \rightarrow \mathbb{R}$ by $g(x) = -\|x\|^2 = -\sum_{i=1}^n x_i^2$.
- The function g is a polynomial in the coordinates of x , hence continuous.
- Rewriting the inequality:

$$\|x\|^2 < 1 \Leftrightarrow -\|x\|^2 > -1 \Leftrightarrow g(x) > -1$$

- Thus, $A = \{x \in \mathbb{R}^n : g(x) > -1\}$. By part (a), A is open.

• **Set $B = \{x \in \mathbb{R}^n : x^T Q x \leq 1\}$**

- Define $h: \mathbb{R}^n \rightarrow \mathbb{R}$ by $h(x) = -x^T Q x = -\sum_{i=1}^n \sum_{j=1}^n q_{ij} x_i x_j$.
- Because $h(x)$ is a quadratic form (a homogeneous polynomial of degree 2), it is continuous on $\mathbb{R}^n$.
- Rewriting the inequality:

$$x^T Q x \leq 1 \Leftrightarrow -x^T Q x \geq -1 \Leftrightarrow h(x) \geq -1$$

- Thus, $B = \{x \in \mathbb{R}^n : h(x) \geq -1\}$. By part (b), B is closed.

• **Set $C = \{(x, y) \in \mathbb{R}^2 : y = x^2\}$**

- Define $k: \mathbb{R}^2 \rightarrow \mathbb{R}$ by $k(x, y) = y - x^2$, which is continuous on $\mathbb{R}^2$.
- Express the equality $y = x^2$ as the intersection of two weak inequalities:

$$y = x^2 \Leftrightarrow (y - x^2 \geq 0) \text{ and } (y - x^2 \leq 0)$$

- Applying this to C :

$$C = \{(x, y) \in \mathbb{R}^2 : k(x, y) \geq 0\}$$

$$\cap \{(x, y) \in \mathbb{R}^2 : -k(x, y) \geq 0\}$$

- By part (b), both $\{(x, y) \in \mathbb{R}^2 : k(x, y) \geq 0\}$ and $\{(x, y) \in \mathbb{R}^2 : -k(x, y) \geq 0\}$ are closed sets.
- Since the intersection of two closed sets is closed, C is closed.

In your mind, make clear the distinct roles played by connectives and clauses

$$\exists \{x_{n_k}\}_{k=1}^{\infty} \subseteq \{x_n\}_n; \quad x_{n_k} \rightarrow x^* \in F$$
$$\{X_n\}_{n=1}^{\infty} \subseteq K \text{ since } F \subseteq K$$
$$\exists \{x_n\}_{n=1}^{\infty} \rightarrow x^* \in K$$
$$\lim_{k \rightarrow \infty} x_{n_k} = x^*$$

Since X_n is convergent
seq contained in F } $\lim_{k \rightarrow \infty} y_k = y^* \in F$
As limit point x^*
must be in F

in F have 9

Subjunctive which

subsequence which
converges to a point in F
+ F must be compact

Suppose $K \subseteq \mathbb{R}^n$ is compact and

$$f: K \rightarrow \mathbb{R}^m$$

is continuous,

- Prove that $f(K)$ is compact.
- Deduce that if $m = 1$, then f attains both a maximum and a minimum on K .
- Construct examples showing that the conclusion in part (b) can fail if:
 - the domain is bounded but not closed;
 - the domain is closed but unbounded;
 - the domain is compact but f is not continuous.

Guidance for part (a). Take a sequence

$$y_k \in f(K).$$

Choose $x_k \in K$ such that

$$f(x_k) = y_k.$$

Now use compactness and continuity in that order.

equivalent to topological compactness defined via open covers as well as to being closed and bounded by the Heine-Borel theorem.)

Guarantee of Convergence Within the Closed Subset

Let $K \subset \mathbb{R}$ be compact, and let $F \subset K$ be a closed subset of $\mathbb{R}$. Let $(x_n)_{n=1}^\infty$ be a sequence in F .

The guarantee that the limit point belongs to F follows from a two-step mechanism: **compactness of the ambient set K** guarantees the existence of a convergent subsequence, and the **closedness of F** guarantees containment of the limit.

from <https://www.google.com/search?q=1866+china+population&rlz=1C1G5e8c3d14162be>

Step-by-Step Analytical Proof

1. Existence of the Subsequential Limit via Compactness:

Since $F \subseteq K$, any sequence $(x_n) \subseteq F$ is also a sequence in the compact set K . By the sequential compactness of K , there exists a subsequence $(x_{n_k})^\infty$ and a point $x \in K$ such that:

$$\lim_{k \rightarrow \infty} x_{n_k} = x$$

2. Containment in the Closed Set via Topological Closure:

A set $F \subseteq \mathbb{R}$ is closed if and only if it contains all of its sequential limit points. That is:

$$\overline{F} = F \Leftrightarrow \left(\forall (y_k) \subseteq F \text{ with } \lim_{k \rightarrow \infty} y_k = y \Rightarrow y \in F \right)$$

Because (x_{n_k}) is entirely contained in F and converges to x , x is a limit point (and closure point) of F . Therefore, $x \in \overline{F} = F$.

Clarification on Arbitrary Sequences vs. Convergent Sequences

- If the sequence (x_n) is already assumed to converge to some x in $\mathbb{R}$:

xin E:
The compactness of K is not even required. The topological closedness of F alone ensures $x \in F$ because:

$$x \in \overline{F} = F$$

- If the sequence (x_n) is arbitrary (not assumed a priori to converge): Compactness of K guarantees the existence of at least one accumulation point x and a convergent subsequence $x_{n_k} \rightarrow x$, while the closedness of F ensures that every such accumulation point resides in F .

From: <<https://remini.google.com/app/480e88c6d141d0be/>>

Proof of Part (a)
Definition: Let $K \subseteq \mathbb{R}^n$ be compact and $f: K \rightarrow \mathbb{R}^m$ be continuous.
 $f(K) \subseteq \mathbb{R}^m$ is compact.

via Sequential Compactness:
 $\{y_k\}_{k=1}^\infty$ be an arbitrary sequence in $f(K)$.

Image Selection: By definition of the image $f(K)$, for each N , there exists $x_k \in K$ such that $f(x_k) = y_k$. This constructs a sequence $\{x_k\}_{k=1}^\infty \subseteq K$.

Sequential Compactness of Domain: Because $K \subseteq \mathbb{R}^n$ is compact, it is sequentially compact. Therefore, there exists a subsequence $\{x_{k_j}\}_{j=1}^\infty \subseteq \{x_k\}_{k=1}^\infty$ and a limit point $x^* \in K$ such that:

$$x_{k_j} \rightarrow x^*$$

Continuity of f at x^* : The function f is continuous on K , hence sequentially continuous at $x^* \in K$. Passing limits through continuous maps yields:

$$f(x_{k_j}) = f\left(\lim_{j \rightarrow \infty} x_{k_j}\right) = f(x^*)$$

Convergence in Image: Define the corresponding subsequence y_{k_j} by $y_{k_j} := f(x_{k_j})$. It follows that:

$$y_{k_j} = f(x^*)$$

Choose $x_k \in K$ such that

$$f(x_k) = y_k.$$

Now use compactness and continuity in that order.

$$\begin{aligned} \exists \{x_n\}_{n=1}^{\infty} : \lim_{n \rightarrow \infty} x_n = x^* \in K \\ \text{Apply } f \text{ to all elements of } \{x_n\}_{n=1}^{\infty} \\ \{f(x_n)\}_{n=1}^{\infty} = \lim_{n \rightarrow \infty} f(x_n) = f(\lim_{n \rightarrow \infty} x_n) \text{ by cont. of } f \\ f(x^*) \in K \end{aligned}$$

$$\begin{aligned} \text{Thus } \lim_{n \rightarrow \infty} f(x_n) = f(x^*) \\ \text{so } \{f(x_n)\}_{n=1}^{\infty} \rightarrow f(x^*) \in K \end{aligned}$$

$f(\{x_n\}_{n=1}^{\infty}) \subseteq K$ Thus for
all convergent sequences in K ,
 $\exists$ convergent subsequence
 $f(\{x_n\}_{n=1}^{\infty})$ w/ limit point
 $f(x^*) \in K$, so K compact

b) if $m=1$ s.t. $f: K \rightarrow \mathbb{R}$
w/ f attains max & min on K
 $f(K)$ is compact by part (a)
since $f(K) \subseteq \mathbb{R}$, by
Heine-Borel $f(K)$ is closed
& bounded s.t. $\exists m, M \in \mathbb{R}$
 $\forall y \in f(K), m \leq y \leq M$

Since $f(K)$ closed, $m, M \in f(K)$

thus f attains max M

& min m on $\mathbb{R}$

c.i) Part (b) can fail if K bounded but
not closed

ex: let $K = (0, 1) \subseteq \mathbb{R}$

$$f(x) = x$$

$$\max f(K) = 1 \text{ but } 1 \notin f(K)$$

Proposition: Let $K \subseteq \mathbb{R}^n$ be compact and $f: K \rightarrow \mathbb{R}$ be continuous (i.e., $m = 1$). Then f attains both a maximum and a minimum on K (The Extreme Value Theorem).

Proof:

1. Compactness of the Image:

By the result of part (a), the image set $f(K) \subseteq \mathbb{R}$ is compact.

2. Heine-Borel Characterization:

By the Heine-Borel Theorem in Euclidean space, $f(K)$ is closed and bounded in $\mathbb{R}$.

3. Existence of Supremum and Infimum:

Because K is non-empty, $f(K)$ is a non-empty, bounded subset of $\mathbb{R}$. By the Completeness Axiom (Least Upper Bound Property) of $\mathbb{R}$, the supremum and infimum exist as finite real numbers:

$$M = \sup f(K) = \sup_{x \in K} f(x), \quad m = \inf f(K) = \inf_{x \in K} f(x)$$

4. Attainment via Closedness:

By definition of the supremum, for every $\varepsilon > 0$, there exists an element $y \in f(K)$ such that $M - \varepsilon < y \leq M$. Hence, M is an adherent point of $f(K)$, meaning $M \in \overline{f(K)}$. Since $f(K)$ is closed, it contains all its accumulation and limit points, so $\overline{f(K)} = f(K)$. Therefore:

$$M \in f(K)$$

By identical reasoning for the infimum, $m \in \overline{f(K)} = f(K)$.

5. Conclusion:

Since $M, m \in f(K)$, there exist points $x_{\max}, x_{\min} \in K$ such that:

$$f(x_{\max}) = M = \sup_{x \in K} f(x) \quad \text{and} \quad f(x_{\min}) = m = \inf_{x \in K} f(x)$$

Thus, f attains its global maximum at $x_{\max}$ and its global minimum at $x_{\min}$ on K . ■

From: <https://www.khanacademy.org/multivariable-calculus/multivariable-derivatives/multivariable-extrema/a/multivariable-extrema-1/a1>

Counterexamples for Part (c)

• (i) Domain is bounded but not closed (K not compact)

◦ Let $K = (0, 1) \subset \mathbb{R}$, which is bounded but open.

◦ Define $f: (0, 1) \rightarrow \mathbb{R}$ by $f(x) = \frac{1}{x}$.

◦ The function f is continuous on $(0, 1)$, but $\sup_{x \in (0, 1)} f(x) = \infty$.

f is unbounded above and fails to attain a maximum.

◦ Bounded alternative: $f(x) = x \sin(1/x)$ is bounded

$\sup f(K) = 1$ and $\inf f(K) = 0$, yet neither the supremum nor the infimum is attained on $(0, 1)$.

$$\begin{array}{ll} \max (f(K)) = 1 & \text{but } 1 \notin f(K) \\ \min (f(K)) = 0 & 0 \notin f(K) \end{array}$$

Ex: 1.11 (i) $\mathbb{R}^n$ is closed but not bounded

ok: Wf $K = [0, \infty)$ & $f(x) = x \cdot \sin(x)$

$K^c = (-\infty, 0)$ is open

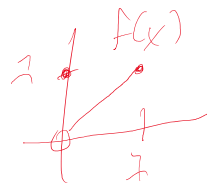
$$\sup (f(K)) = \infty \quad \text{but } \sup \text{ \& min do not exist}$$

(iii) $\text{Aut}(G)$ sub. A K compact
but f not cont

ex let $K = [0, 1]$ & $f(x) = \begin{cases} 1 & \text{if } x=0 \\ x & \text{if } x \neq 0 \end{cases}$

ind $f(k) = 0$

$$\text{WA } 0 \notin f(K)$$



f has f never
a f has its min

Problem 6. Continuous bijections from compact sets

Let $K \subseteq \mathbb{R}^n$ be compact, and suppose

$$f: K \rightarrow Y \subseteq \mathbb{R}^m$$

is a continuous bijection,

Prove that

$$f^{-1}: Y \rightarrow K$$

is continuous.

Guided contradiction argument. Take a sequence

$$y_k \rightarrow y$$

and define

$$x_k = f^{-1}(y_k), \quad x = f^{-1}(y),$$

Suppose that $x_k \not\rightarrow x$. Show that there exist $\varepsilon > 0$ and a subsequence (x_{k_j}) such that

II. — II. —

$$K \subseteq \mathbb{R}^n \text{ compact}$$
$$f: K^m \rightarrow \mathbb{R}^m$$

is bijection

- (ii) **Domain is closed but unbounded (K not compact)**
 - Let $K = [0, \infty) \subset \mathbb{R}$, which is closed in $\mathbb{R}$ but unbounded.
 - Define $f: [0, \infty) \rightarrow \mathbb{R}$ by $f(x) = x$.
 - The function f is continuous on $[0, \infty)$, but $\sup_{x \in [0, \infty)} f(x) = \infty$ and it attains no maximum.
 - **Bounded alternative:** $f(x) = 1 - e^{-x}$ on $[0, \infty)$ is bounded and continuous with $\sup_{x \geq 0} f(x) = 1$, but $f(x) < 1$ for all x and the maximum is never attained.

- (iii) Domain is compact but f is not continuous
 - Let $K = [0,1] \subset \mathbb{R}$, which is compact (closed and b
 - Define $f: [0,1] \rightarrow \mathbb{R}$ by:

$$f(x) = \begin{cases} \frac{1}{x} & \text{if } x \in (0,1] \\ 0 & \text{if } x = 0 \end{cases}$$

- f is discontinuous at $x = 0$, unbounded on $[0,1]$, and has no maximum.
- *Bounded alternative:* Define $g: [0,1] \rightarrow \mathbb{R}$ by:

$$g(x) = \begin{cases} x & \text{if } x \in [0,1) \\ 0 & \text{if } x = 1 \end{cases}$$

• g is bounded with $\sup_{x \in [0,1]} g$

- g is bounded with $\sup_{x \in [0,1]} g(x) = 1$, but $g(x) < 1$ for
 so g attains no maximum on K .

act)
unded.
 $(x) = \infty$, so
unded and
 $\in [0, \infty)$, so

ounded).

nd has no

all $x \in [0,1]$.

Problem 6: Continuous Bijections from Compact Sets
Proposition: Let $K \subseteq \mathbb{R}^n$ be compact, $Y \subseteq \mathbb{R}^m$, and suppose $f: K \rightarrow Y$ is a continuous bijection. Then the inverse map $f^{-1}: Y \rightarrow K$ is continuous.

Proof via Guided Contradiction:
To show that $f^{-1}: Y \rightarrow K$ is continuous, it suffices to prove sequential continuity on Y .

Let $(y_k)_{k=1}^\infty \subseteq Y$ be an arbitrary sequence converging to $y \in Y$, i.e., $y_k \rightarrow y$.

Define:

$$x_k := f^{-1}(y_k) \in K \quad \text{and} \quad x := f^{-1}(y) \in K$$

and define $y_k \rightarrow y$

$$x_k = f^{-1}(y_k), \quad x = f^{-1}(y).$$

Suppose that $x_k \not\rightarrow x$. Show that there exist $\epsilon > 0$ and a subsequence (x_{k_j}) such that

$$\|x_{k_j} - x\| \geq \epsilon$$

for every j .

Use compactness to extract a further convergent subsequence

$$x_{k_{j_l}} \rightarrow x^*.$$

Now compare

$$f(x_{k_{j_l}}) \rightarrow f(x^*)$$

with

$$f(x_{k_{j_l}}) = y_{k_{j_l}} \rightarrow y.$$

Identify where uniqueness of limits is used and where injectivity of f is used.

cont bijection

WTS $f^{-1}: Y \rightarrow K$

is continuous

f^{-1} is continuous at point x^*

i.e. every seq. $\{x_n\}_{n=1}^{\infty}$ converging to x^* maps to seq. $\{f^{-1}(x_n)\}_{n=1}^{\infty}$ converging to $f^{-1}(x^*)$

let $y_k \rightarrow y$ be a seq. in Y

s.t. $x_k = f^{-1}(y_k) \neq x = f^{-1}(y)$

for contradiction, suppose $x_k \not\rightarrow x$

since $\{x_k\}_{k=1}^{\infty} \subseteq K$ is a seq. in K & K compact,

$\exists \{x_{k_j}\}_{j=1}^{\infty} \rightarrow x$ a convergent subseq. in K

thus $\lim_{j \rightarrow \infty} x_{k_j} = x$ implies that $\forall \epsilon > 0$,

$\|x_{k_j} - x\| \geq \epsilon$ for all $j \in \mathbb{N}$

since x_{k_j} is itself a seq. there must exist another convergent subseq. $x_{k_{j_l}} \subseteq K$

s.t. $x_{k_{j_l}} \rightarrow x^* \neq x$

We wish to show that $x_k \rightarrow x$.

1. **Negation of Convergence:**

Assume, for contradiction, that $x_k \nrightarrow x$.
The definition of convergence states: $\forall \varepsilon > 0, \exists N \in \mathbb{N}$ such that $\forall k \geq N, \|x_k - x\| < \varepsilon$.

2. Negating this statement yields:

$$\exists \varepsilon_0 > 0 \quad \text{such that} \quad \forall N \in \mathbb{N}, \exists k \geq N \text{ with } \|x_k - x\| \geq \varepsilon_0$$

This condition allows the inductive construction of a strictly increasing sequence of indices $k_1 < k_2 < k_3 < \dots$ defining a subsequence $(x_{k_j})_{j=1}^\infty$ such that:

$$\|x_{k_j} - x\| \geq \varepsilon_0 \quad \forall j \in \mathbb{N}$$

3. **Extraction of a Convergent Subsequence:**

Because $(x_{k_j})_{j=1}^\infty \subseteq K$ and K is compact (hence sequentially compact), there exists a further subsequence $(x_{k_{j_\ell}})_{\ell=1}^\infty$ that converges to some limit $x^* \in K$:

$$\lim_{\ell \rightarrow \infty} x_{k_{j_\ell}} = x^*$$

4. **Comparison of Limits in the Image:**

- **Via Continuity of f :** Since f is continuous on K , it is sequentially continuous at x^* . Therefore:

$$\lim_{\ell \rightarrow \infty} f(x_{k_{j_\ell}}) = f(x^*)$$

- **Via the Original Convergent Sequence:** By definition, $f(x_{k_{j_\ell}}) = y_{k_{j_\ell}}$. Since $(y_{k_{j_\ell}})_{\ell=1}^\infty$ is a subsequence of the convergent sequence $y_k \rightarrow y$, every subsequence must converge to the same limit:

$$\lim_{\ell \rightarrow \infty} f(x_{k_{j_\ell}}) = \lim_{\ell \rightarrow \infty} y_{k_{j_\ell}} = y$$

5. **Applying Uniqueness of Limits and Injectivity:**

- **Uniqueness of Limits in $\mathbb{R}^m$:** Because $\mathbb{R}^m$ is a metric space (Hausdorff), limit points of convergent sequences are unique. Equating the two limits gives:
 $f(x^*) = y$
- **Injectivity of f :** We already have $f(x) = y$. Thus $f(x^*) = f(x)$. Because f is injective (since f is a bijection):

$$x^* = x$$

6. **Contradiction:**

Since $x_{k_{j_\ell}} \rightarrow x^* = x$, it follows by the definition of the limit that:

$$\lim_{\ell \rightarrow \infty} \|x_{k_{j_\ell}} - x\| = 0$$

However, by construction, every element of the initial subsequence satisfies:

$$\|x_{k_{j_\ell}} - x\| \geq \varepsilon_0 > 0 \quad \forall \ell \in \mathbb{N}$$

Taking the limit as $\ell \rightarrow \infty$ yields $0 \geq \varepsilon_0 > 0$, a contradiction.

Thus, the assumption that $x_k \nrightarrow x$ is false. It follows that $x_k \rightarrow x$, which implies $\lim_{k \rightarrow \infty} f^{-1}(y_k) = f^{-1}(y)$. Therefore, f^{-1} is sequentially continuous, and hence continuous on Y . ■

From <<https://gemini.google.com/app/06b7ab97dac5cbbb>>

on is
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s.t. $x_{k_{i_2}} \rightarrow x$
 since f is not injective

$$f(x_{k_{i_2}}) \rightarrow f(x^*)$$

each term in $x_{k_{i_2}}$ maps to
 same $y_{k_{i_2}} \in Y$ & $f(x^*) = y$

$$\text{thrs } f^{-1}(y) = x \text{ & } f^{-1}(y_{k_{i_2}}) = x_{k_{i_2}}$$

& $x_{k_{i_2}} \rightarrow x \in R$, thrs the

seq. $x_k \rightarrow x$ since $x_{k_{i_2}} \subseteq x_k$

but by assumption $x_k \not\rightarrow x$, contradiction

so $x_k \not\rightarrow x$ must be false

& f^{-1} must be continuous

3 Differentiability as Linear Approximation

Problem 7. The derivative is unique

Suppose $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfies

$$f(x+h) = f(x) + Ah + r_A(h)$$

and also

$$f(x+h) = f(x) + Bh + r_B(h),$$

where $A, B: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are linear maps and

$$\frac{\|r_A(h)\|}{\|h\|} \rightarrow 0, \quad \frac{\|r_B(h)\|}{\|h\|} \rightarrow 0.$$

Prove that

$$A = B.$$

Guidance. Fix an arbitrary $v \in \mathbb{R}^n$ and set

$$h = tv.$$

Compare the two approximations and divide by $t \neq 0$. Then let $t \rightarrow 0$.

Why does showing

$$(A - B)v = 0$$

for every v imply $A = B$?

7) Derivative is unique

Suppose $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is defined as

$$f(x+h) = f(x) + Ah + r_A(h)$$

$$\text{& } f(x+h) = f(x) + Bh + r_B(h)$$

where $A, B: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are linear maps

$$\text{& } \frac{\|r_A(h)\|}{\|h\|} \rightarrow 0 \quad \frac{\|r_B(h)\|}{\|h\|} \rightarrow 0$$

on is
 effective
 ed?

21

Problem 7: The Derivative is Unique

Proposition: Suppose $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ satisfies

$f(x+h) = f(x) + Ah + r_A(h)$ and $f(x+h) = f(x) + Bh + r_B(h)$
 where $A, B: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are linear maps and $\lim_{h \rightarrow 0} \frac{\|r_A(h)\|}{\|h\|} = 0$, $\lim_{h \rightarrow 0} \frac{\|r_B(h)\|}{\|h\|} = 0$. Then $A = B$.

Proof:

1. Equating the Linear Approximations:

Subtracting the two expressions for $f(x+h)$ gives:

$$(f(x) + Ah + r_A(h)) - (f(x) + Bh + r_B(h)) = 0$$

$$(A - B)h = r_B(h) - r_A(h)$$

2. Evaluating Along a Direction:

Fix an arbitrary vector $v \in \mathbb{R}^n$.

- If $v = 0$, then by linearity $(A - B)0 = 0$.
- If $v \neq 0$, let $h = tv$ for a scalar $t \in \mathbb{R} \setminus \{0\}$.
 Substituting $h = tv$ into the equation and utilizing the linearity of $(A - B)$:

$$(A - B)(tv) = r_B(tv) - r_A(tv)$$

$$t(A - B)v = r_B(tv) - r_A(tv)$$

3. Taking the Limit as $t \rightarrow 0$:

Dividing both sides by $t \neq 0$ and applying the Euclidean norm on $\mathbb{R}^m$:

$$\|(A - B)v\| = \left\| \frac{r_B(tv) - r_A(tv)}{t} \right\|$$

Applying the triangle inequality:

$$\|(A - B)v\| \leq \frac{\|r_B(tv)\|}{|t|} + \frac{\|r_A(tv)\|}{|t|}$$

Since $\|tv\| = |t|\|v\|$, we rewrite this in terms of the remainder limits by multiplying and dividing by $\|v\| > 0$:

$$\|(A - B)v\| < \|v\| \left(\frac{\|r_B(tv)\|}{\|tv\|} + \frac{\|r_A(tv)\|}{\|tv\|} \right)$$

$$\frac{\epsilon}{\|h\|} \geq \frac{\epsilon}{\|h\|}$$

Wts that $A=B$ must be true

let $v \in \mathbb{R}^n$ be arbitrary vector

& define $h = \epsilon v$ for $\epsilon \in \mathbb{R}$
s.t. $\epsilon \neq 0$

Assume $A \neq B$ for contradiction

$$f(x+h) = f(x) + A h + r_A(h) = f(x) + A \epsilon v + r_A(\epsilon v)$$

$$\text{dividing by } \epsilon: \frac{f(x+\epsilon v)}{\epsilon} = \frac{f(x)}{\epsilon} + A v + \frac{r_A(\epsilon v)}{\epsilon}$$

Since $A \neq B$,

$$A v \neq B v$$

$$A \epsilon v \neq B \epsilon v$$

$$f(x) + A \epsilon v \neq f(x) + B \epsilon v$$

$$\frac{f(x)}{\epsilon} + A v \neq \frac{f(x)}{\epsilon} + B v$$

$$\lim_{\epsilon \rightarrow 0} \frac{\|r_A(\epsilon v)\|}{\|\epsilon v\|} = 0$$

$$\text{so } \frac{f(x+\epsilon v)}{\epsilon} =$$

$$\frac{f(x+\epsilon v)}{\epsilon} =$$

$$\text{thus } \frac{f(x)}{\epsilon} + A v$$

hA this is
assumption
... ..

$$\|tv\| = |t| \|v\|$$

Since $\|tv\| = |t| \|v\|$, we rewrite this in terms of the remainder limits by multiplying and dividing by $\|v\| > 0$:

$$\|(A - B)v\| \leq \|v\| \left(\frac{\|r_B(tv)\|}{\|tv\|} + \frac{\|r_A(tv)\|}{\|tv\|} \right)$$

As $t \rightarrow 0$, $h = tv \rightarrow 0$. Taking the limit of both sides as $t \rightarrow 0$:

$$\|(A - B)v\| \leq \|v\| \left(\lim_{t \rightarrow 0} \frac{\|r_B(tv)\|}{\|tv\|} + \lim_{t \rightarrow 0} \frac{\|r_A(tv)\|}{\|tv\|} \right) = \|v\|(0 + 0) = 0$$

Since the norm is non-negative, $\|(A - B)v\| = 0$, which implies:

$$(A - B)v = 0$$

Why $(A - B)v = 0$ for all v implies $A = B$

• **Definition of Operator Equality:** Two functions (or linear transformations)

$A, B: \mathbb{R}^n \rightarrow \mathbb{R}^m$ are defined to be equal if and only if they agree on their entire domain:

$$Av = Bv \quad \forall v \in \mathbb{R}^n$$

Since $(A - B)v = 0 \Leftrightarrow Av = Bv$ holds for every arbitrary $v \in \mathbb{R}^n$, the linear operators A and B are identical.

• **Basis Representation:** In terms of standard coordinates, if $(A - B)e_j = 0$ for each standard basis vector $e_j \in \mathbb{R}^n$ ($j = 1, \dots, n$), every column of the matrix representation of $(A - B)$ is the zero vector in $\mathbb{R}^m$, establishing that $A - B = 0_{m \times n} \Rightarrow A = B$. ■

From <<https://gemini.google.com/app/06b7ab97dac5cbb>>

$$\frac{r_A(tv)}{t}$$

$$\frac{\|r_A(tv)\|}{\|tv\|} \rightarrow 0$$

$$\frac{f(x)}{t} + Av$$

$$\frac{f(x)}{t} + Bv \quad \text{by same steps}$$

$$Av = \frac{f(x)}{t} + Bv$$

$$\text{contradicts our } A \neq B, A = B$$

$$Av = Bv$$

$$Av - Bv = 0$$

$$(A - B)v = 0$$

assumption

this is a function

Problem 8. Directional derivatives are not enough

Consider

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0), \\ 0, & (x, y) = (0, 0). \end{cases}$$

(a) Prove that f is continuous at $(0, 0)$.

(b) Compute

$$f_x(0, 0), \quad f_y(0, 0).$$

(c) For an arbitrary direction $v = (a, b)$, compute

$$D_v f(0, 0).$$

(d) Show that every directional derivative at the origin exists.

(e) Is the map

$$v \mapsto D_v f(0, 0)$$

linear?

(f) Deduce that f is not differentiable at the origin.

Conclude by explaining, in your own words, the distinction among

partial derivatives, directional derivatives, the derivative $Df(x)$.

8)

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases} \quad f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

1) wts f is continuous at $(0, 0)$
if f is cont, then $\forall \epsilon > 0$,

$\exists \delta > 0$ s.t. if $|(x, y) - (0, 0)| < \delta$

$$\lim |f(x, y) - f(0, 0)| < \epsilon$$

$$f(x, y) = 0 \quad \text{pick } \delta = (\epsilon, \epsilon)$$

$$\text{if } |(0, 0) - (0, 0)| < \delta = (\epsilon, \epsilon)$$

$$(0, 0) < \delta = (\epsilon, \epsilon)$$

$$\text{then } |f(0, 0) - f(0, 0)| < \epsilon$$

$$0 < \epsilon$$

$$L = (0, 0) \quad \text{since} \quad \lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 y}{x^2 + y^2} = 0$$

by L'Hopital's

b) $f_x(0, 0)$ partial derivative

$$f_x(x, y) = \begin{cases} \frac{2xy(x^2 + y^2) - (x^2)(x^2 y)}{(x^2 + y^2)^2} = \frac{2x^3 y + 2xy^3 - x^4 y}{(x^2 + y^2)^2} = \frac{2xy^3}{(x^2 + y^2)^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

$H \neq B,$
 $\downarrow$
 $A = B \quad \square$

$\rightarrow \parallel$

$(0,0)$

$\uparrow,$

(ε)

(ε)

$(0,0)$

we

$\nabla f \neq (0,0)$

$$\frac{f'}{g} = \frac{f'g - g'f}{g^2}$$

Critique of Handwritten Attempt

- **Invalid Choice of δ :** In a metric space $(\mathbb{R}^2, \|\cdot\|)$, the radius δ must be a strictly positive real scalar ($\delta > 0$), not a 2D vector like $(\varepsilon, \varepsilon)$.
- **Circular / Trivialized Evaluation:** Substituting $(x, y) = (0, 0)$ directly into the distance condition reduces the statement to testing $|0 - 0| < \varepsilon$, which only checks that $f(0, 0)$ equals itself. A continuity proof must establish that $|f(x, y) - f(0, 0)| < \varepsilon$ for **all** (x, y) in a punctured neighborhood $0 < \sqrt{x^2 + y^2} < \delta$.
- **Misapplication of L'Hôpital's Rule:** L'Hôpital's rule is valid only for single-variable limits. Multivariable limits $\lim_{(x,y) \rightarrow (0,0)}$ cannot be evaluated by taking standard single-variable quotients of derivatives.
- **Target Value Conflation:** The function maps to $\mathbb{R} (m = 1)$, so the limit value is the scalar $L = 0$, not the vector $(0, 0)$.

Formal Proof of Part (a)

Proposition: The function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$.

Proof via ε - δ and Squeeze Argument:

To show continuity at $(0, 0)$, we must prove:

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = f(0, 0) = 0$$

which requires showing that for every $\varepsilon > 0$, there exists $\delta > 0$ such that:

$$0 < \|(x, y) - (0, 0)\| = \sqrt{x^2 + y^2} < \delta \Rightarrow |f(x, y) - 0| < \varepsilon$$

1. Bounding the Function:

For any $(x, y) \neq (0, 0)$, observe that $x^2 \leq x^2 + y^2$ (since $y^2 \geq 0$). Thus:

$$\frac{x^2}{x^2 + y^2} \leq 1$$

2. Estimating the Error:

Using this bound:

$$|f(x, y) - 0| = \left| \frac{x^2 y}{x^2 + y^2} \right| = \frac{x^2}{x^2 + y^2} |y| \leq 1 \cdot |y| = |y|$$

Furthermore, $|y| = \sqrt{y^2} \leq \sqrt{x^2 + y^2} = \|(x, y)\|$. Therefore:

$$|f(x, y)| \leq \|(x, y)\|$$

3. Selection of δ :

Given any $\varepsilon > 0$, choose $\delta = \varepsilon$.

If $0 < \|(x, y)\| < \delta$, then:

$$|f(x, y) - 0| \leq \|(x, y)\| < \delta = \varepsilon$$

From <https://www.geogebra.org/m/5a97708/2020/04/04/2020/04/04>

To evaluate partial and directional derivatives at a piecewise point like $(0, 0)$, do not differentiate the algebraic expression for $(x, y) \neq (0, 0)$ via the quotient rule. Instead, apply the **limit definition of the derivative** directly at the base point.

Part (b): Compute $f_x(0, 0)$ and $f_y(0, 0)$

Using the difference quotient definition of partial derivatives at $(0, 0)$:

- **Partial with respect to x :**

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(0 + h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{f(h, 0) - 0}{h}$$

For $h \neq 0$, the function value along the x -axis is:

$$f(h, 0) = \frac{h^2 \cdot 0}{h^2 + 0^2} = 0$$

$$f_x(0,0) = 0$$

$$f_y(0,0) = 0 \text{ by symmetry}$$

c) for arbitrary $v = (a,b)$

$$D_v f(0,0) = (a,b) ?$$

d) WTS every directional derivative
at the origin exists

isn't that just
part (c)?

e) let $v \rightarrow D_v f(0,0)$

be a map Γ from
arbitrary vector v to
the directional derivative
at the vector at the
origin

Γ is not a linear mapping
since ...

f) f is not differentiable
at the origin since

g) partial derivatives are the approx
to the change in a function

Therefore:

$$f_x(0,0) = \lim_{h \rightarrow 0} \frac{0}{h} = \lim_{h \rightarrow 0} 0 = 0$$

• **Partial with respect to y :**

$$f_y(0,0) = \lim_{k \rightarrow 0} \frac{f(0,0+k) - f(0,0)}{k} = \lim_{k \rightarrow 0} \frac{f(0,k) - 0}{k}$$

For $k \neq 0$, the function value along the y -axis is:

$$f(0,k) = \frac{0^2 \cdot k}{0^2 + k^2} = 0$$

Therefore:

$$f_y(0,0) = \lim_{k \rightarrow 0} \frac{0}{k} = \lim_{k \rightarrow 0} 0 = 0$$

$$\mathbf{f}_x(0,0) = \mathbf{0}, \quad \mathbf{f}_y(0,0) = \mathbf{0}$$

Part (c): Compute $D_v f(0,0)$ for an arbitrary direction $v = (a,b)$

Let $v = (a,b) \in \mathbb{R}^2$ be an arbitrary vector. The directional derivative at $(0,0)$ in direction v is given by:

$$D_v f(0,0) = \lim_{t \rightarrow 0} \frac{f((0,0) + t(a,b)) - f(0,0)}{t} = \lim_{t \rightarrow 0} \frac{f(ta, tb)}{t}$$

• **Case 1:** $v = (0,0)$

$$D_{(0,0)} f(0,0) = \lim_{t \rightarrow 0} \frac{f(0,0)}{t} = 0$$

• **Case 2:** $v = (a,b) \neq (0,0)$

For $t \neq 0$, $(ta, tb) \neq (0,0)$. Evaluating $f(ta, tb)$:

$$f(ta, tb) = \frac{(ta)^2(tb)}{(ta)^2 + (tb)^2} = \frac{t^3 a^2 b}{t^2(a^2 + b^2)} = t \frac{a^2 b}{a^2 + b^2}$$

Substituting this into the limit:

$$D_v f(0,0) = \lim_{t \rightarrow 0} \frac{t \frac{a^2 b}{a^2 + b^2}}{t} = \lim_{t \rightarrow 0} \frac{a^2 b}{a^2 + b^2} = \frac{a^2 b}{a^2 + b^2}$$

Combining both cases (since $\frac{a^2 b}{a^2 + b^2}$ evaluates to 0 if $a = 0$ or $b = 0$):

$$D_v \mathbf{f}(0,0) = \begin{cases} \mathbf{a}^2 \mathbf{b} \\ \mathbf{a}^2 + \mathbf{b}^2 \end{cases}, \quad (\mathbf{a}, \mathbf{b}) \neq (0,0)$$

$$\mathbf{0}, \quad (\mathbf{a}, \mathbf{b}) = (0,0)$$

Part (d): Show that every directional derivative at the origin exists

- For the zero vector $v = (0,0)$, $D_{(0,0)} f(0,0) = 0 \in \mathbb{R}$.
- For every non-zero direction $v = (a,b) \in \mathbb{R}^2 \setminus \{(0,0)\}$, $a^2 + b^2 = \|v\|^2 > 0$. The expression $\frac{a^2 b}{a^2 + b^2}$ is a well-defined real number with no dependence on t .

Because the difference quotient limit converges to a finite real number for every vector $v \in \mathbb{R}^2$, the directional derivative $D_v f(0,0)$ exists for all directions $v \in \mathbb{R}^2$.

From <https://www.youtube.com/watch?v=950Cz0dUx04>.

Part (e): Linearity of the Directional Derivative Map

The map $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $T(v) = D_v f(0,0)$ is **not linear**.

To verify linearity, T must satisfy additivity: $T(u + v) = T(u) + T(v)$ for all $u, v \in \mathbb{R}^2$.

- Let $u = (1,0)$ and $w = (0,1)$.

$$T(u) = D_{(1,0)} f(0,0) = \frac{1^2 \cdot 0}{1^2 + 0^2} = 0$$

$$T(w) = D_{(0,1)} f(0,0) = \frac{0^2 \cdot 1}{0^2 + 1^2} = 0$$

$$T(u) + T(w) = 0 + 0 = 0$$

- Consider the sum $u + w = (1,1)$:

$$T(u + w) = D_{(1,1)} f(0,0) = \frac{1^2 \cdot 1}{1^2 + 1^2} = \frac{1}{2}$$

Because $T(u + w) = \frac{1}{2} \neq 0 = T(u) + T(w)$, additivity fails, and the map $v \mapsto D_v f(0,0)$ is non-linear.

Part (f): Deducing Non-Differentiability at the Origin

Theorem: If a function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at x_0 , then for every direction $v \in \mathbb{R}^n$, the directional derivative $D_v f(x_0)$ exists and satisfies:

$$D_v f(x_0) = Df(x_0) \cdot v$$

where $Df(x_0)$ is the total derivative (a linear transformation / Jacobian matrix). Consequently, the map $v \mapsto D_v f(x_0)$ must be linear.

- Deduction:** From Part (e), the directional derivative operator $v \mapsto D_v f(0,0)$ is non-linear. By contraposition, f cannot be differentiable at $(0,0)$.
- Direct Verification via Remainder:** If f were differentiable at $(0,0)$, its total derivative candidate would be $A = \nabla f(0,0) = (f_x(0,0), f_y(0,0)) = (0,0)$. The linear approximation error is:

$$r(h) = f(h_1, h_2) - f(0,0) - Ah = \frac{h_1^2 h_2}{h_1^2 + h_2^2} - 0 - 0 = \frac{h_1^2 h_2}{h_1^2 + h_2^2}$$

Taking $h = (t, t)$ along the diagonal as $t \rightarrow 0^+$:

$$\frac{\|r(h)\|}{\|h\|} = \frac{\frac{t^3}{2t^2}}{\sqrt{t^2 + t^2}} = \frac{\frac{t}{2}}{t\sqrt{2}} = \frac{1}{2\sqrt{2}} \neq 0$$

Since $\lim_{h \rightarrow 0} \frac{\|r(h)\|}{\|h\|} \neq 0$, f is not differentiable at $(0,0)$.

y) partial derivatives = rate of the change in a fuct in the direction of an axis according to the basis

directional der. are approximations of the fuct in any arbitrary direction, not just axes

the total derivative $DA(x)$ describes a linear approximation in any direction

Problem 9. Continuous partial derivatives imply differentiability

Suppose

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

has partial derivatives in a neighborhood of (a, b) , and suppose f_x and f_y are continuous at (a, b) .

Prove that f is differentiable at (a, b) .

Begin by writing

$$f(a+h, b+k) - f(a, b) = [f(a+h, b+k) - f(a, b+k)] + [f(a, b+k) - f(a, b)].$$

1. Apply the one-dimensional mean value theorem to the two differences.

2. Show that, for suitable intermediate points ξ and η ,

$$f(a+h, b+k) - f(a, b) = f_x(\xi, b+k)h + f_y(a, \eta)k.$$

3. Subtract

$$f_x(a, b)h + f_y(a, b)k.$$

4. Bound the absolute value of the resulting remainder by

$$|f_x(\xi, b+k) - f_x(a, b)| |h| + |f_y(a, \eta) - f_y(a, b)| |k|.$$

5. Divide by

$$\sqrt{h^2 + k^2}$$

and complete the proof.

Indicate precisely where continuity of the partial derivatives is used.

4) Cont partial derivatives imply differentiability

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

has partial derivatives in neighb of (a, b)

& f_x, f_y cont at (a, b)

wts f is differentiable at (a, b)

for small perturbation of (a, b)

$$\text{by } (h, k) \in \mathbb{R}^2,$$

$$f(a+h, b+k) - f(a, b) = [f(a+h, b+k) - f(a, b+k)] + [f(a, b+k) - f(a, b)]$$

$$\frac{\|r(h)\|}{\|h\|} = \frac{\frac{t^3}{2t^2}}{\sqrt{t^2 + t^2}} = \frac{\frac{t}{2}}{t\sqrt{2}} = \frac{1}{2\sqrt{2}} \neq 0$$

Since $\lim_{h \rightarrow 0} \frac{\|r(h)\|}{\|h\|} \neq 0$, f is not differentiable at $(0,0)$.

Distinction Among Partial Derivatives, Directional Derivatives, and the Derivative $Df(x)$

- **Partial Derivatives** ($f_{x_i}(x)$): 1-dimensional rates of change restricted exclusively along the standard coordinate axes (the basis vectors e_i). They provide purely axis-aligned information and ignore behavior along all other trajectories.
- **Directional Derivatives** ($D_v f(x)$): 1-dimensional rates of change along an arbitrary straight-line vector v . While they capture directional variation along all straight lines through x , they remain isolated 1D limits and do not enforce any uniform coordination across different directions.
- **The Derivative** ($Df(x)$): A global, n -dimensional **linear map** (represented by the Jacobian matrix) that provides a uniform first-order tangent hyperplane approximation to the graph:

$$\lim_{h \rightarrow 0} \frac{\|f(x+h) - f(x) - Df(x)h\|}{\|h\|} = 0$$

Differentiability requires that the error vanishes uniformly from *all* paths approaching x (curved or straight), binding all directional derivatives together into a single coherent linear operator.

From <<https://gemini.google.com/app/06b7ab97dac5cbbb>>

The equation decomposes a 2D diagonal displacement (h, k) into two 1D, axis-aligned steps:

$$f(a+h, b+k) - f(a, b) = [f(a+h, b+k) - f(a, b+k)] + [f(a, b+k) - f(a, b)]$$

This is powerful because the **1D Mean Value Theorem (MVT)** can only be applied when one variable changes at a time. Breaking the joint increment into sequential 1-variable steps converts the 2D problem into two standard single-variable calculus problems, directly linking the global increment to partial derivatives.

Step-by-Step Proof of Problem 9

Proposition: If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ has partial derivatives in an open neighborhood U of (a, b) , and f_x, f_y are continuous at (a, b) , then f is differentiable at (a, b) .

Proof:

1. Applying the 1D Mean Value Theorem:

Choose $(h, k) \neq (0,0)$ sufficiently small such that the rectangular path from (a, b) to $(a+h, b+k)$ remains entirely within U .

- Define the single-variable function $g(t) = f(t, b+k)$ on the interval between a and $a+h$. Since g is differentiable on this interval, the 1D MVT guarantees the existence of an intermediate point ξ strictly between a and $a+h$ such that:

$$f(a+h, b+k) - f(a, b+k) = g(a+h) - g(a) = g'(\xi)h = f_x(\xi, b+k)h$$

- Define $u(s) = f(a, s)$ on the interval between b and $b+k$. By the 1D MVT, there exists an intermediate point η strictly between b and $b+k$ such that:

$$f(a, b+k) - f(a, b) = u(b+k) - u(b) = u'(\eta)k = f_y(a, \eta)k$$

2. Summing the Increments:

Substituting these into the identity:

$$f(a+h, b+k) - f(a, b) = f_x(\xi, b+k)h + f_y(a, \eta)k$$

3. Defining the Remainder Term:

To establish differentiability with candidate linear map $A = (f_x(a, b) \quad f_y(a, b))$, subtract $f_x(a, b)h + f_y(a, b)k$ from both sides:

$$r(h, k) = f(a+h, b+k) - f(a, b) - [f_x(a, b)h + f_y(a, b)k]$$

$$r(h, k) = [f_x(\xi, b+k) - f_x(a, b)]h + [f_y(a, \eta) - f_y(a, b)]k$$

4. Bounding the Remainder:

Taking the absolute value and applying the triangle inequality:

$$|r(h, k)| \leq |f_x(\xi, b+k) - f_x(a, b)||h| + |f_y(a, \eta) - f_y(a, b)||k|$$

5. Dividing by $\|(h, k)\| = \sqrt{h^2 + k^2}$ and Taking the Limit:

Since $|h| \leq \sqrt{h^2 + k^2}$ and $|k| \leq \sqrt{h^2 + k^2}$, we have $\frac{|h|}{\sqrt{h^2 + k^2}} \leq 1$ and $\frac{|k|}{\sqrt{h^2 + k^2}} \leq 1$:

$$\frac{|r(h, k)|}{\sqrt{h^2 + k^2}} \leq |f_x(\xi, b+k) - f_x(a, b)| + |f_y(a, \eta) - f_y(a, b)|$$

As $(h, k) \rightarrow (0,0)$:

- $\xi \rightarrow a$ and $(b+k) \rightarrow b \Rightarrow (\xi, b+k) \rightarrow (a, b)$
- $\eta \rightarrow b \Rightarrow (a, \eta) \rightarrow (a, b)$

Where continuity is used:

Because f_x and f_y are continuous at (a, b) :

$+ [f(a, b+k) - f(a, b)]$
 Applying mean value theorem in each dim
 if $f(a, b)$ cont. then $\forall a_0, a_1$ with $a_0 \leq a_1$
 if $a_* \in [a_0, a_1]$,
 $f(a_*) \in [f(a_0), f(a_1)]$
 holding b constant
 thus for intermediate points ξ & η
 $f(a+h, b+k) - f(a, b) = f_x(\xi, b)h + f_y(a, \eta)k$

4 Restricting Multivariate Problems to Lines

Problem 10. The multivariate mean-value idea

Let

$$f: U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$$

be differentiable, and suppose the entire line segment joining a and b lies in U .

Define

$$g(t) = f(a + t(b - a)), \quad 0 \leq t \leq 1.$$

(a) Prove that

$$g'(t) = Df(a + t(b - a))(b - a).$$

Approach 1: Using the Chain Rule

Define the affine curve $\gamma: [0, 1] \rightarrow \mathbb{R}^n$ by:

$$\gamma(t) = a + t(b - a)$$

• The map γ is differentiable on $(0, 1)$ with derivative (velocity vector):

$$\gamma'(t) = b - a$$

• Since $g(t) = (f \circ \gamma)(t)$ and f is differentiable on U , the multivariate chain rule applies directly:

$$g'(t) = Df(\gamma(t)) \cdot \gamma'(t) = Df(a + t(b - a))(b - a)$$

Approach 2: From First Principles (Linear Approximation / Remainder Definition)

Fix $t \in (0, 1)$ and let $x(t) = a + t(b - a)$. For any non-zero scalar increment Δt , the perturbed point is:

$$x(t + \Delta t) = a + (t + \Delta t)(b - a) = x(t) + \Delta t(b - a)$$

Let $h = \Delta t(b - a) \in \mathbb{R}^n$. Because f is differentiable at $x(t)$, we expand $f(x(t) + h)$ using its linear approximation

$$f(x(t) + h) - f(x(t)) = Df(x(t))h + r(h)$$

where $\lim_{h \rightarrow 0} \frac{|r(h)|}{\|h\|} = 0$.

1. **Substitute** $h = \Delta t(b - a)$:

$$g(t + \Delta t) - g(t) = Df(x(t))(\Delta t(b - a)) + r(\Delta t(b - a))$$

2. **Apply Linearity of the Operator** $Df(x(t))$:

$$g(t + \Delta t) - g(t) = \Delta t \cdot Df(x(t))(b - a) + r(\Delta t(b - a))$$

3. **Form the Difference Quotient:**

Divide through by $\Delta t \neq 0$:

$$\frac{g(t + \Delta t) - g(t)}{\Delta t} = Df(x(t))(b - a) + \frac{r(\Delta t(b - a))}{\Delta t}$$

4. **Evaluate the Limit as $\Delta t \rightarrow 0$:**

Examine the error term:

$$\left| \frac{r(\Delta t(b - a))}{\Delta t} \right| = \|b - a\| \cdot \frac{|r(\Delta t(b - a))|}{\|\Delta t(b - a)\|}$$

Since $\Delta t \rightarrow 0 \Rightarrow h = \Delta t(b - a) \rightarrow 0$, the differentiability of f ensures:

$$\lim_{\Delta t \rightarrow 0} \frac{|r(\Delta t(b - a))|}{\|\Delta t(b - a)\|} = 0$$

Therefore:

$$g'(t) = \lim_{\Delta t \rightarrow 0} \frac{g(t + \Delta t) - g(t)}{\Delta t} = Df(x(t))(b - a) = Df(a + t(b - a))(b - a) \quad \blacksquare$$

From <<https://gemini.google.com/app/06b7ab97dac5cbbb>>

(b) Apply the one-dimensional mean value theorem to show that there exists a point c on the segment from a to b such that

$$f(b) - f(a) = Df(c)(b - a).$$

Part (b): Mean Value Theorem on the Segment

1. **Check MVT Hypotheses:**

- The single-variable function $g(t) = f(a + t(b - a))$ is continuous on $[0, 1]$ (composition of continuous maps).
- From part (a), g is differentiable on $(0, 1)$.

2. **Apply 1D Mean Value Theorem:**

There exists some $\tau \in (0, 1)$ such that:

$$F(a, b)]$$

ension

$$\begin{matrix} \nearrow f \\ \downarrow \\ + \\ \downarrow \\ + \end{matrix}$$

$$)h + f_y(a, \eta)k$$

- As $(h, k) \rightarrow (0, 0)$:
- $\xi \rightarrow a$ and $(b + k) \rightarrow b \Rightarrow (\xi, b + k) \rightarrow (a, b)$
 - $\eta \rightarrow b \Rightarrow (a, \eta) \rightarrow (a, b)$

Where continuity is used:

Because f_x and f_y are continuous at (a, b) :

$$\lim_{(h,k) \rightarrow (0,0)} |f_x(\xi, b + k) - f_x(a, b)| = 0 \quad \text{and} \quad \lim_{(h,k) \rightarrow (0,0)} |f_y(a, \eta) - f_y(a, b)| = 0$$

Therefore:

$$\lim_{(h,k) \rightarrow (0,0)} \frac{|r(h, k)|}{\sqrt{h^2 + k^2}} = 0$$

This satisfies the definition of total differentiability, proving f is differentiable at (a, b) with derivative $Df(a, b) = (f_x(a, b) \quad f_y(a, b))$. ■

From <<https://gemini.google.com/app/06b7ab97dac5cbbb>>

(c) Deduce the bound

$$|f(b) - f(a)| \leq \sup_{z \in [a, b]} \|Df(z)\| \|b - a\|.$$

$g(1) - g(0) = g'(c)(1 - 0)$
3. Substitute Definitions:
 $\circ g(1) = f(a + 1(b - a)) = f(b)$
 $\circ g(0) = f(a + 0(b - a)) = f(a)$
 $\circ$ From part (a), $g'(c) = Df(a + \tau(b - a))(b - a)$
4. Define the Intermediate Point:
Set $c = a + \tau(b - a)$. Since $\tau \in (0, 1)$, c is a point lying strictly on the open line segment joining a and b (denoted $c \in (a, b)$).
Substituting these in yields:

$$f(b) - f(a) = Df(c)(b - a)$$

Part (c): Deducing the Upper Bound

1. Take the Absolute Value:
From part (b), for the point $c \in [a, b]$:

$$|f(b) - f(a)| = |Df(c)(b - a)|$$

2. Apply the Operator Norm Inequality:
For any linear map $A: \mathbb{R}^n \rightarrow \mathbb{R}$ and vector $v \in \mathbb{R}^n$, the operator norm $\|A\| = \sup_{\|u\|=1} |Au|$ satisfies:

$$|Av| \leq \|A\| \|v\|$$

Applying this with $A = Df(c)$ and $v = b - a$:

$$|Df(c)(b - a)| \leq \|Df(c)\| \|b - a\|$$

3. Bound by the Supremum Over the Segment:
Because $c \in [a, b]$, the quantity $\|Df(c)\|$ is bounded by the supremum over all points z on the line segment $[a, b]$:

$$\|Df(c)\| \leq \sup_{z \in [a, b]} \|Df(z)\|$$

Combining these inequalities gives:

$$|f(b) - f(a)| \leq \sup_{z \in [a, b]} \|Df(z)\| \|b - a\| \quad \blacksquare$$

From <https://www.geogebra.com/m/qy067xap7fay5dddy>.

(d) Suppose now that U is convex and

$$Df(x) = 0 \quad \text{for every } x \in U.$$

Prove that f is constant on U .

Proof technique to remember. A useful strategy in multivariate calculus is

multivariate problem $\longrightarrow$ restrict to a line $\longrightarrow$ apply one-variable calculus.

To establish the claim:

- Let $a, b \in U$ be arbitrary points.
- Because U is convex, the entire line segment $[a, b] = \{a + t(b - a) : t \in [0, 1]\}$ is contained in U .
- Since f is differentiable on U , it is differentiable at every point along the segment $[a, b]$. By Part (b), there exists some intermediate point $c \in (a, b) \subseteq U$ such that:

$$f(b) - f(a) = Df(c)(b - a)$$

- By assumption, $Df(x) = 0$ (the zero linear operator) for every $x \in U$. Since $c \in U$, we have $Df(c) = 0$.
- Applying the zero transformation to $(b - a)$ gives:

$$f(b) - f(a) = 0(b - a) = 0 \implies f(b) = f(a)$$

- Since a and b were chosen arbitrarily in U , fixing a base point $x_0 \in U$ yields $f(x) = f(x_0)$ for all $x \in U$. Hence, f is constant on U .

From <https://www.geogebra.com/m/qy067xap7fay5dddy>.

Problem 11. If the Hessian vanishes, the function is affine

Let $U \subseteq \mathbb{R}^n$ be open and convex. Suppose

$$f: U \rightarrow \mathbb{R}$$

is C^2 and

$$H_f(x) = 0 \quad \text{for every } x \in U.$$

Prove that there exist $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$ such that

$$f(x) = a^\top x + b \quad \text{for every } x \in U.$$

Guidance. Treat

$$\nabla f: U \rightarrow \mathbb{R}^n$$

as a function in its own right.

What is

$$D(\nabla f)(x)?$$

Use the previous problem to show that ∇f is constant. Then consider

$$g(x) = f(x) - a^\top x.$$

To solve Problem 11, apply the result of Problem 10(d) twice in succession: first to the vector-valued gradient map ∇f , and second to the scalar-valued adjusted function $g(x)$.

Step 1: Relate the Hessian to the Derivative of the Gradient

Define the gradient mapping:

$$F: U \rightarrow \mathbb{R}^n, \quad F(x) = \nabla f(x) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(x) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x) \end{pmatrix}$$

Because f is C^2 , each coordinate function $F_i(x) = \frac{\partial f}{\partial x_i}(x)$ is C^1 and therefore differentiable on U . The total derivative of this vector-valued function F is its Jacobian matrix, which is precisely the Hessian of f :

$$D(\nabla f)(x) = J_F(x) = H_f(x)$$

By assumption, $H_f(x) = 0_{n \times n}$ for all $x \in U$. Applying Problem 10(d) component-wise to each F_i (or directly to the vector map on the convex domain U) gives:

$D(F_i)(x) = 0 \implies F_i(x)$ is constant on $U \quad \forall i = 1, \dots, n$
Thus, the entire gradient vector is constant across U :

$$\nabla f(x) = a \quad \text{for some fixed vector } a \in \mathbb{R}^n$$

Step 2: Construct the Auxiliary Function $g(x)$
Consider the function suggested in the guidance:

$$g: U \rightarrow \mathbb{R}, \quad g(x) = f(x) - a^\top x$$

Compute the total derivative (or gradient) of g :

$$Dg(x) = Df(x) - a^\top$$

In gradient terms:

$$\nabla g(x) = \nabla f(x) - a$$

Since $\nabla f(x) = a$ for all $x \in U$:

$$\nabla g(x) = a - a = 0 \quad \forall x \in U$$

Step 3: Apply Problem 10(d) to $g(x)$

Since U is convex and $Dg(x) = 0$ everywhere on U , Problem 10(d) implies that $g(x)$ must be constant on U :

$g(x) = b$ for some constant scalar $b \in \mathbb{R}$
Substituting back into the definition of $g(x)$:

$f(x) - a^\top x = b \implies f(x) = a^\top x + b \quad \forall x \in U \quad \blacksquare$

From <<https://gemini.google.com/app/06b7ab97dac5cbbb>>

Problem 12. Taylor’s theorem in several variables

Let

$f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$

be C^2 , and suppose that the line segment from x to $x + h$ lies entirely in U .

Define

$g(t) = f(x + th), \quad 0 \leq t \leq 1.$

(a) Show that

$g'(t) = \nabla f(x + th)^\top h.$

(b) Show that

$g''(t) = h^\top H_f(x + th)h.$

(c) Apply the one-dimensional Taylor theorem to g between 0 and 1 to prove that, for some $\theta \in (0, 1)$,

$f(x + h) = f(x) + \nabla f(x)^\top h + \frac{1}{2}h^\top H_f(x + \theta h)h.$

Why does the second-order term involve the scalar quadratic form

$h^\top H_f h$

rather than merely the vector $H_f h$?

Why the Gradient and Hessian Act on h

The perturbation $h \in \mathbb{R}^n$ is not acting as an affine coordinate translation; it represents the **fixed direction and displacement vector** parameterizing the line segment $\gamma(t) = x + th$.

- **First Derivative / Linear Term:** Differentiating along the line $\gamma(t)$ projects the gradient ∇f onto the velocity vector $\gamma'(t) = h$, producing the scalar directional rate of change:

$g'(t) = \nabla f(\gamma(t))^\top h$

- **Second Derivative / Quadratic Term:** Differentiating $g'(t)$ a second time measures how this directional rate of change varies as you continue along that exact same vector h :

$g''(t) = \frac{d}{dt} \left[\sum_{i=1}^n \frac{\partial f}{\partial x_i}(x + th) h_i \right] = \sum_{i=1}^n \sum_{j=1}^n \frac{\partial^2 f}{\partial x_j \partial x_i}(x + th) h_j h_i = h^\top H_f(x + th)h$

Step-by-Step Proof of Problem 12

Part (a): Compute $g'(t)$

Define $\gamma: [0, 1] \rightarrow U$ by $\gamma(t) = x + th$. Then $\gamma'(t) = h$.
By the chain rule:

$g'(t) = Df(\gamma(t))\gamma'(t) = \nabla f(x + th)^\top h = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(x + th) h_i$

Part (b): Compute $g''(t)$

Differentiating $g'(t)$ with respect to t :

$g''(t) = \frac{d}{dt} \left(\sum_{i=1}^n \frac{\partial f}{\partial x_i}(x + th) h_i \right)$

Applying the chain rule to each component map $x \mapsto \frac{\partial f}{\partial x_i}(x)$:

$g''(t) = \sum_{i=1}^n \left(\sum_{j=1}^n \frac{\partial^2 f}{\partial x_j \partial x_i}(x + th) h_j \right) h_i = \sum_{i=1}^n \sum_{j=1}^n h_i \left[H_f(x + th) \right]_{ij} h_j$

In compact matrix notation:

$g''(t) = h^\top H_f(x + th)h$

Part (c): Apply 1D Taylor’s Theorem to g

The single-variable function $g: [0, 1] \rightarrow \mathbb{R}$ is C^2 on $[0, 1]$. By the single-variable Taylor Theorem with Lagrange remainder expanded around $t = 0$ and evaluated at $t = 1$, there exists some $\theta \in (0, 1)$ such that:

$g(1) = g(0) + g'(0)(1 - 0) + \frac{1}{2}g''(\theta)(1 - 0)^2$

Substituting the expressions for $g(t)$, $g'(t)$, and $g''(t)$:

- $g(1) = f(x + h)$
- $g(0) = f(x)$
- $g'(0) = \nabla f(x)^\top h$
- $g''(\theta) = h^\top H_f(x + \theta h)h$

Yields the multivariate Taylor expansion:

$f(x + h) = f(x) + \nabla f(x)^\top h + \frac{1}{2}h^\top H_f(x + \theta h)h \quad \blacksquare$

Why the Second-Order Term is the Scalar Quadratic Form $h^\top H_f h$ rather than $H_f h$

- **Dimensional Consistency:** The function $f: U \rightarrow \mathbb{R}$ is scalar-valued. Thus, the quantity $f(x + h) - f(x)$ is a scalar in $\mathbb{R}$. The first-order term $\nabla f(x)^\top h$ is an inner product (scalar).
 - $H_f(x)$ is an $n \times n$ matrix (the Hessian operator).
 - The product $H_f h$ is an $n \times 1$ vector (measuring the rate of change of the gradient vector ∇f). You cannot add a vector $H_f h \in \mathbb{R}^n$ to a scalar $f(x) \in \mathbb{R}$.
- **Bilinear Form Mechanics:** The second total differential $D^2 f(x)$ is intrinsically a symmetric bilinear form $D^2 f(x): \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$. Evaluating this bilinear form along the perturbation pair (h, h) yields the quadratic form $D^2 f(x)(h, h) = h^\top H_f(x)h \in \mathbb{R}$.

From <<https://gemini.google.com/app/06b7ab97dac5cbbb>>

5 Inverse and Implicit Functions

Problem 13. Local versus global invertibility

Consider

$F(x, y) = \begin{pmatrix} e^x \cos y \\ e^x \sin y \end{pmatrix}.$

Problem 13: Local Versus Global Invertibility

Let $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by:

$F(x, y) = \begin{pmatrix} F_1(x, y) \\ F_2(x, y) \end{pmatrix} = \begin{pmatrix} e^x \cos y \\ e^x \sin y \end{pmatrix}$

(Note: Identifying $\mathbb{R}^2 \cong \mathbb{C}$ with $z = x + iy$, this mapping represents the complex exponential $f(z) = e^z = e^x(\cos y + i \sin y)$).

Part (a): Compute $DF(x, y)$

The total derivative matrix (Jacobian) is formed by the partial derivatives of the coordinate functions:

Dimension Reduction (nD → 1D) Proving Taylor's Theorem directly in n dimensions requires juggling multivariable limits and mixed partial derivatives simultaneously. The parameterized line $\gamma(t) = x + t\mathbf{h}$ restricts the n-dimensional scalar field $f: U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ to a 1-dimensional slice, producing a standard single-variable function $g = f \circ \gamma: [0,1] \rightarrow \mathbb{R}$.

Using Single-Variable Theorems: Once the problem is reduced to the single-variable function $g(t)$, you can immediately invoke classical 1D theorems—such as the 1D Mean Value Theorem and 1D Taylor Theorem with Lagrange remainder—which are already proven from single-variable real analysis.

Practical Application of the Chain Rule: Viewing $\gamma(t)$ as an explicit curve $\gamma: [0,1] \rightarrow U$ allows you to apply the multivariate Chain Rule cleanly:

$$D(f \circ \gamma)(t) = Df(\gamma(t)) \cdot \gamma'(t)$$

Since $\gamma'(t) = \frac{d}{dt}(x + t\mathbf{h}) = \mathbf{h}$ is the constant velocity vector of the path, the Chain Rule systematically pulls out the direction $\mathbf{h}$ at each derivative order.

<https://gemini.google.com/app/06b7ab97dac5cbbb>

The Core Philosophy: Local Invertibility of Nonlinear Maps

At this level, the **Inverse Function Theorem (IFT)** and the **Implicit Function Theorem** address a fundamental question: *When does the local behavior of a nonlinear mapping mirror the behavior of its linear approximation (the derivative operator)?*

If a linear transformation $T: \mathbb{R}^n \rightarrow \mathbb{R}^n$ (represented by an $n \times n$ matrix A) has $\det(A) \neq 0$, then T is a global bijection with a continuous linear inverse T^{-1} . The IFT generalizes this principle locally to C^1 nonlinear mappings.

The Inverse Function Theorem (IFT)

Statement: Let $U \subseteq \mathbb{R}^n$ be an open set, and let $f: U \rightarrow \mathbb{R}^n$ be a C^1 mapping. Suppose at a point $x_0 \in U$, the total derivative operator:

Inverse Function Theorem (IFT)

• **Setting & Core Condition:** Let $f: U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^1 on open U . If $x_0 \in U$ satisfies:

$$\det Df(x_0) \neq 0 \quad (\text{or } Df(x_0) \text{ is invertible})$$

then there exist open neighborhoods $V \ni x_0$ and $W \ni f(x_0)$ such that $f: V \rightarrow W$ is a C^1 -diffeomorphism.

• **Inverse Derivative Formula:** For $y \in W$ and $x = f^{-1}(y) \in V$:

Problem 13. Local versus global invertibility

Consider

F(x,y)=(e^x cos y)/(e^x sin y).

- (a) Compute DF(x,y).
- (b) Show that

det DF(x,y) ≠ 0

- for every (x,y) ∈ ℝ².
- (c) State precisely what the inverse function theorem implies at an arbitrary point (x,y).
- (d) Show that

F(x,y+2π)=F(x,y).

- (e) Is F globally one-to-one? Explain why your answer does not contradict the inverse function theorem.

Now consider

f(x)=x³.

- (f) The function is globally one-to-one, but

f'(0)=0.

- Why does this not contradict the inverse function theorem?
- (g) Write down f⁻¹ and determine whether f⁻¹ is differentiable at 0.

Your answer should make clear the distinctions between

local and global

and between

sufficient and necessary conditions.

F(x,y)=(F1(x,y))/(F2(x,y))=(e^x cos y)/(e^x sin y)

(Note: Identifying ℝ² ≃ ℂ with z = x + iy, this mapping represents the complex exponential f(z) = e^z = e^x(cos y + i sin y)).

Part (a): Compute DF(x,y)

The total derivative matrix (Jacobian) is formed by the partial derivatives of the coordinate functions:

DF(x,y)=(∂F1/∂x ∂F1/∂y)/(∂F2/∂x ∂F2/∂y)=(e^x cos y -e^x sin y)/(e^x sin y e^x cos y)

Part (b): Show that det DF(x,y) ≠ 0 for every (x,y) ∈ ℝ²

Compute the determinant:

det DF(x,y)=(e^x cos y)(e^x cos y) - (-e^x sin y)(e^x sin y)

det DF(x,y)=e^2x cos² y + e^2x sin² y = e^2x(cos² y + sin² y) = e^2x

Since the exponential function satisfies e^2x > 0 for all real x ∈ ℝ:

det DF(x,y)=e^2x > 0 ∀(x,y) ∈ ℝ²

Thus, det DF(x,y) ≠ 0 everywhere on ℝ².

Part (c): Precise Implication of the Inverse Function Theorem

Because F ∈ C¹(ℝ²) and det DF(x₀,y₀) ≠ 0 at every point (x₀,y₀) ∈ ℝ², the Inverse Function Theorem guarantees that:

For every (x₀,y₀) ∈ ℝ², there exists an open neighborhood U ⊂ ℝ² containing (x₀,y₀) and an open neighborhood V ⊂ ℝ² containing F(x₀,y₀) such that the restricted map:

F|U: U → V

is a C¹-diffeomorphism (a bijective mapping with a continuously differentiable inverse (F|U)⁻¹: V → U).

Part (d): Show that F(x,y+2π) = F(x,y)

Using the 2π-periodicity of sine and cosine:

F(x,y+2π)=(e^x cos(y+2π))/(e^x sin(y+2π))=(e^x cos y)/(e^x sin y)=F(x,y)

Part (e): Global Injectivity and Consistency with the IFT

- Global Injectivity: F is not globally one-to-one (not injective). By Part (d), distinct points in ℝ² that differ by integer multiples of 2π in their second coordinate map to the exact same output:

F(0,0)=(1/0)=F(0,2π), yet (0,0) ≠ (0,2π)

- Resolution / Consistency with the IFT: This does not contradict the Inverse Function Theorem. The Inverse Function Theorem is strictly a local result. It guarantees the existence of small open neighborhoods (e.g., of radius smaller than π) on which F is locally one-to-one. Non-zero Jacobian determinant everywhere det DF(x) ≠ 0 is a necessary condition for local invertibility, but it is not sufficient to guarantee global invertibility across the entire domain.

From <https://gemini.google.com/app/06b7ab97dac5cbbb>

Part (f): Why f'(0) = 0 for f(x) = x³ Does Not Contradict the IFT

The function f: ℝ → ℝ defined by f(x) = x³ is strictly increasing, bijective, and hence globally one-to-one, yet its derivative is f'(x) = 3x² = f'(0) = 0.

This does not contradict the Inverse Function Theorem because:

- The IFT is a Conditional Guarantee (Sufficient, Not Necessary for Invertibility): The theorem states that if f'(x₀) ≠ 0, then f has a local C¹ inverse near x₀. It does not state that non-zero derivative is necessary for a function to be invertible in a purely set-theoretic (or continuous) sense.
- The IFT Pertains to Smooth (C¹) Inverses: What fails when f'(0) = 0 is not the existence of a local inverse, but rather the differentiability of that inverse at the image point f(0) = 0.

Part (g): Inverse Function and Differentiability at 0

- The Inverse Function: Solving y = x³ for x yields:

f⁻¹(y) = y¹/³ (or f⁻¹(x) = x¹/³)

f⁻¹ is continuous everywhere on ℝ.

- Differentiability at 0: Evaluating the difference quotient for f⁻¹ at y = 0:

lim y→0 (f⁻¹(y) - f⁻¹(0))/(y - 0) = lim y→0 (y¹/³ - 0)/y = lim y→0 1/y²/³ = ∞

Because this limit diverges to +∞, f⁻¹ is not differentiable at 0 (the graph of f⁻¹ exhibits a vertical tangent at the origin).

Synthesis: Distinctions Between Local vs. Global and Sufficient vs. Necessary

- Local vs. Global:
 - Local Properties: The condition det DF(x) ≠ 0 describes the infinitesimal behavior of F in an arbitrarily small neighborhood of a point. By Problem 13(a–e), having det DF(x) ≠ 0 everywhere guarantees that the function is a local diffeomorphism around every point, but fails to prevent distinct, distant regions of the domain from mapping to the same output (e.g., 2π-periodicity in the complex exponential). Global invertibility requires additional topological conditions (such as properness and simple connectivity via Hadamard's Global Inversion Theorem).
 - Global Properties: A function can be globally one-to-one over its entire domain (like f(x) = x³) even if its derivative degenerates at isolated critical points.
- Sufficient vs. Necessary Conditions:
 - Sufficient for a C¹ Local Inverse: det DF(x₀) ≠ 0 is a sufficient condition to guarantee that a local inverse exists and is continuously differentiable (C¹).
 - Not Necessary for Topological Invertibility: det DF(x₀) ≠ 0 is not necessary for a function to have a continuous local inverse f⁻¹ (as demonstrated by f(x) = x³, where f⁻¹(y) = y¹/³ is a well-defined homeomorphism).
 - Necessary for a Differentiable Inverse: Differentiating the identity f⁻¹(f(x)) = x via the chain rule yields D(f⁻¹)(f(x)) · Df(x) = I. Taking determinants gives det D(f⁻¹)(f(x)) · det Df(x) = 1. Hence, det Df(x₀) ≠ 0 is strictly necessary if the inverse map f⁻¹ is to be differentiable at f(x₀).

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Problem 14: What Can Be Solved for Implicitly?

Consider the function F(x,y,z) = x² + y² + z² - 1 at the point p = (0,0,1) on the unit sphere S². The gradient of F at an arbitrary point is:

∇F(x,y,z)=(Fx,Fy,Fz)=(2x,2y,2z)

Evaluating at p = (0,0,1):

∇F(0,0,1)=(0,0,2)

Part (a): Solving Locally in the Form z = g(x,y)

- Answer: Yes, the equation can locally be solved for z as a C¹ function z = g(x,y) near (0,0).

- Determining Partial Derivative: The relevant partial derivative is Fz(x,y,z) = ∂F/∂z.

- Justification:

- F(0,0,1) = 0² + 0² + 1² - 1 = 0.
- Fz(0,0,1) = 2(1) = 2 ≠ 0.

By the Implicit Function Theorem, because the partial derivative with respect to the target variable z is non-

Problem 14. What can be solved for implicitly?

Consider the level surface

F(x,y,z)=x²+y²+z²-1=0.

First consider the point

p=(0,0,1).

- (a) Can the equation locally be solved in the form

z=g(x,y)?

Which partial derivative determines the answer?

The Inverse Function Theorem (IFT)

Statement:
Let $U \subseteq \mathbb{R}^n$ be an open set, and let $f: U \rightarrow \mathbb{R}^n$ be a C^1 mapping. Suppose at a point $x_0 \in U$, the total derivative operator:

$Df(x_0): \mathbb{R}^n \rightarrow \mathbb{R}^n$
is invertible (i.e., the Jacobian determinant $\det J_f(x_0) \neq 0$).

- Then:
- Local Diffeomorphism:** There exists an open neighborhood $V \subseteq U$ containing x_0 and an open neighborhood $W \subseteq \mathbb{R}^n$ containing $f(x_0)$ such that $f: V \rightarrow W$ is a **bijective homeomorphism** (one-to-one, onto, with continuous inverse).
 - Differentiability of the Inverse:** The inverse mapping $f^{-1}: W \rightarrow V$ is also C^1 .
 - Derivative of the Inverse:** For every $y \in W$ with $y = f(x)$, the derivative of the inverse is the matrix inverse of the forward derivative:

$D(f^{-1})(y) = [Df(x)]^{-1}$
Proof Mechanism (The Contraction Mapping Principle):
To solve $f(x) = y$ locally for x , define the auxiliary operator:

- $\phi_y(x) = x + [Df(x_0)]^{-1}(y - f(x))$
- A fixed point $\phi_y(x) = x$ occurs if and only if $f(x) = y$.
 - The derivative $D\phi_y(x) = I - [Df(x_0)]^{-1}Df(x)$. Since Df is continuous, $D\phi_y(x_0) = 0$, so $\|D\phi_y(x)\| \leq \frac{1}{2}$ in a sufficiently small closed ball around x_0 .
 - By the Mean Value Inequality, ϕ_y is a strict contraction on a complete metric space $(\bar{B}(x_0, r))$, guaranteeing a unique local solution $x = f^{-1}(y)$ via the **Banach Fixed-Point Theorem**.

The Implicit Function Theorem

While the IFT handles invertible maps between spaces of the same dimension ($n \rightarrow n$), the Implicit Function Theorem handles underdetermined nonlinear systems $F(x, y) = 0$, where $F: \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^k$. It determines when a system of k equations can be locally solved explicitly for k endogenous variables in terms of n exogenous parameters.

Statement:
Let $U \subseteq \mathbb{R}^n \times \mathbb{R}^k$ be open, and let $F: U \rightarrow \mathbb{R}^k$ be a C^1 mapping. Denote points in U as (x, y) with $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^k$. Suppose at $(x_0, y_0) \in U$:

- $F(x_0, y_0) = 0$
- The partial Jacobian with respect to the y -variables, $D_y F(x_0, y_0) \in \mathcal{L}(\mathbb{R}^k, \mathbb{R}^k)$, is **invertible** (i.e., $\det \left[\frac{\partial F_i}{\partial y_j}(x_0, y_0) \right] \neq 0$).

Then:

- Existence of Local Explicit Function:** There exists an open neighborhood $V \subseteq \mathbb{R}^n$ of x_0 , an open neighborhood $W \subseteq \mathbb{R}^k$ of y_0 , and a unique C^1 mapping $g: V \rightarrow W$ such that:

- $g(x_0) = y_0 \quad \text{and} \quad F(x, g(x)) = 0 \quad \forall x \in V$
- Total Derivative of the Implicit Solution:** Differentiating the identity $F(x, g(x)) = 0$ via the multivariate chain rule gives:

$D_x F(x, g(x)) + D_y F(x, g(x)) \cdot Dg(x) = 0$

Since $D_y F(x, g(x))$ remains invertible near (x_0, y_0) :

$Dg(x) = -\left[D_y F(x, g(x))\right]^{-1} D_x F(x, g(x))$

Structural Comparison		
Feature	Inverse Function Theorem	Implicit Function Theorem
Domain & Codomain	$f: \mathbb{R}^n \rightarrow \mathbb{R}^n$	$F: \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^k$
Core Condition	$\det Df(x_0) \neq 0$	$\det D_y F(x_0, y_0) \neq 0$
Geometric Output	Local diffeomorphism (coordinate chart transformation)	Level set $\{F(x, y) = 0\}$ is locally the graph of a C^1 manifold $y = g(x)$
Derivative Formula	$D(f^{-1})(y) = [Df(x)]^{-1}$	$Dg(x) = -\left[D_y F\right]^{-1} D_x F$

The two theorems are mathematically equivalent: the Implicit Function Theorem is proved by augmenting F into an $(n + k) \rightarrow (n + k)$ map $H(x, y) = (x, F(x, y))$ and applying the Inverse Function Theorem.

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then there exist open neighborhoods $V \ni x_0$ and $W \ni f(x_0)$ such that $f: V \rightarrow W$ is a C^1 -diffeomorphism.

- Inverse Derivative Formula:** For $y \in W$ and $x = f^{-1}(y) \in V$:

$D(f^{-1})(y) = [Df(x)]^{-1} = \left[Df\left(f^{-1}(y)\right)\right]^{-1}$

- Determinant Relation:**

$\det D\left(f^{-1}\right)(y)=\frac{1}{\det Df(x)}$

Implicit Function Theorem (Implicit Function Form)

- System Setup:** Let $F: U \subseteq \mathbb{R}^n \times \mathbb{R}^k \rightarrow \mathbb{R}^k$ be C^1 , with $x \in \mathbb{R}^n$ (exogenous parameters) and $y \in \mathbb{R}^k$ (endogenous variables). Suppose $F(x_0, y_0) = 0$.

- Rank / Invertibility Condition:** The partial Jacobian with respect to y must be non-singular:

$\det D_y F(x_0, y_0) = \det \left[\frac{\partial F_i}{\partial y_j}(x_0, y_0) \right]_{k \times k} \neq 0$

- Implicit Map Existence:** There exists an open set $V \ni x_0$ and a unique C^1 map $g: V \rightarrow \mathbb{R}^k$ such that $g(x_0) = y_0$ and $F(x, g(x)) = 0$ for all $x \in V$.

- Master Derivative Formula ($k \times n$ Jacobian Matrix):**

$Dg(x) = -\left[D_y F(x, g(x))\right]^{-1} D_x F(x, g(x))$

Theoretical Traps & Boundary Rules

- Local $\neq$ Global:** $\det Df(x) \neq 0$ everywhere ensures local invertibility around every point, but does **not** guarantee global injectivity (e.g., $F(x, y) = (e^x \cos y, e^x \sin y)$).

- Sufficiency vs. Necessity:** $\det Df(x_0) \neq 0$ is sufficient for a *differentiable* inverse, but not necessary for a continuous topological inverse (e.g., $f(x) = x^3$).

From <<https://gemini.google.com/app/06b7ab97dac5cbbb>>

(a) Can the equation locally be solved in the form

$$z = g(x, y)?$$

Which partial derivative determines the answer?

(b) Can it locally be solved in the form

$$x = h(y, z)?$$

Explain.

Now consider the point

$$q = (1, 0, 0).$$

(c) Repeat parts (a)–(b) at q .

(d) Explain geometrically why the answers change as we move from p to q .

(e) Suppose that near p ,

$$z = g(x, y).$$

Starting from

$$F(x, y, g(x, y)) = 0,$$

derive formulas for

$$g_x(x, y), \quad g_y(x, y)$$

using the chain rule.

Do not begin by quoting a memorized implicit differentiation formula.

- **Determining Partial Derivative:** The relevant partial derivative is $F_z(x, y, z) = \frac{\partial F}{\partial z}$.
 - **Justification:**
 - $F(0, 0, 1) = 0^2 + 0^2 + 1^2 - 1 = 0$.
 - $F_z(0, 0, 1) = 2(1) = 2 \neq 0$.
- By the Implicit Function Theorem, because the partial derivative with respect to the target variable z is non-zero (invertible in $\mathbb{R}^1$), there exists an open neighborhood $V \subseteq \mathbb{R}^2$ containing $(0, 0)$ and a unique C^1 function $g: V \rightarrow \mathbb{R}$ such that $g(0, 0) = 1$ and $F(x, y, g(x, y)) = 0$.
- (Explicitly: $g(x, y) = \sqrt{1 - x^2 - y^2}$ on the open disk $x^2 + y^2 < 1$.)

Part (b): Solving Locally in the Form $x = h(y, z)$

- **Answer: No,** the equation cannot locally be solved for x as a single-valued function $x = h(y, z)$ in any neighborhood of $(0, 0, 1)$.
- **Explanation via the IFT Condition:**
To solve for x implicitly, the IFT requires the partial derivative with respect to x to be non-zero:

$$F_x(0, 0, 1) = 2(0) = 0$$

Since the linear approximation condition fails, the IFT does not guarantee a local solution.

- **Geometric / Analytical Failure:**
At the "north pole" $p = (0, 0, 1)$, the tangent plane to the sphere is horizontal ($z = 1$). Solving $F(x, y, z) = 0$ for x requires:

$$x^2 = 1 - y^2 - z^2 \Rightarrow x = \pm\sqrt{1 - y^2 - z^2}$$

In any open neighborhood of $(y, z) = (0, 1)$:

- If $z > 1$ (points strictly above the north pole), $1 - y^2 - z^2 < 0$, so there are **no real solutions** for x .
- If $z < 1$ with $y^2 + z^2 < 1$, there are **two distinct solutions** $x = +\sqrt{1 - y^2 - z^2}$ and $x = -\sqrt{1 - y^2 - z^2}$ for every single pair (y, z) .

Because every neighborhood contains inputs (y, z) that produce multiple outputs or no output at all, the relation fails the vertical line test in the x -direction and cannot be represented locally as a well-defined function $x = h(y, z)$.

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Problem 14 (Continued)

Let $F(x, y, z) = x^2 + y^2 + z^2 - 1$. At the point $q = (1, 0, 0)$:

$$\nabla F(1, 0, 0) = \begin{pmatrix} F_x(1, 0, 0) \\ F_y(1, 0, 0) \\ F_z(1, 0, 0) \end{pmatrix} = \begin{pmatrix} 2(1) \\ 2(0) \\ 2(0) \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$$

Part (c): Repeat Parts (a)–(b) at $q = (1, 0, 0)$

- **Solving in the form $z = g(x, y)$ at q :**
 - **Answer: No,** it cannot be solved locally for z .
 - **Reason:** $F_z(1, 0, 0) = 0$. The partial derivative with respect to z vanishes, so the Implicit Function Theorem fails. In any neighborhood of $(x, y) = (1, 0)$, points where $x^2 + y^2 > 1$ give no real solutions for z , while $x^2 + y^2 < 1$ yields two solutions $z = \pm\sqrt{1 - x^2 - y^2}$.
- **Solving in the form $x = h(y, z)$ at q :**
 - **Answer: Yes,** it can be solved locally for x .
 - **Reason:** $F_x(1, 0, 0) = 2 \neq 0$. By the Implicit Function Theorem, there exists an open neighborhood $W \subseteq \mathbb{R}^2$ containing $(0, 0)$ and a unique C^1 function $h: W \rightarrow \mathbb{R}$ such that $h(0, 0) = 1$ and $F(h(y, z), y, z) = 0$. (Explicitly, $h(y, z) = \sqrt{1 - y^2 - z^2}$.)

Part (d): Geometric Explanation of Why the Answers Change

The normal vector to the level surface S^2 is $\mathbf{n} = \nabla F$. The orientation of the tangent plane determines which coordinate axes can serve as independent variables:

- **At $p = (0, 0, 1)$:** The normal vector is $\nabla F = (0, 0, 2)^T$ (parallel to the z -axis). The tangent plane is horizontal ($z = 1$, parallel to the xy -plane). Because the tangent plane projects bijectively onto the xy -plane, the surface can be represented as a graph over (x, y) , so $z = g(x, y)$ is well-defined. Conversely, the tangent plane is perpendicular to the x -axis, meaning the surface "folds over" in the x -direction, causing the vertical line test along x to fail.
- **At $q = (1, 0, 0)$:** The normal vector is $\nabla F = (2, 0, 0)^T$ (parallel to the x -axis). The tangent plane is vertical ($x = 1$, parallel to the yz -plane). This plane projects bijectively onto the yz -plane, allowing the surface to be parameterized as a graph $x = h(y, z)$, but it folds over relative to the z -axis ($F_z = 0$).

Part (e): Deriving $g_x(x, y)$ and $g_y(x, y)$ from First Principles via the Chain Rule

Let $z = g(x, y)$ satisfy the identity:

$$\Phi(x, y) = F(x, y, g(x, y)) \equiv 0$$

for all (x, y) in a neighborhood of $(0, 0)$.

Define the vector-valued map $\psi(x, y) = \begin{pmatrix} x \\ y \\ g(x, y) \end{pmatrix}$, so $\Phi(x, y) = (F \circ \psi)(x, y) = 0$.

- **Differentiating with respect to x :**

Differentiate both sides of $F(x, y, g(x, y)) = 0$ with respect to x using the multivariable chain rule:

$$\frac{\partial}{\partial x} [F(x, y, g(x, y))] = \frac{\partial F}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial g}{\partial x} = 0$$

Since $\frac{\partial x}{\partial x} = 1$ and $\frac{\partial y}{\partial x} = 0$:

$$F_x(x, y, g(x, y)) \cdot 1 + F_y(x, y, g(x, y)) \cdot 0 + F_z(x, y, g(x, y)) g_x(x, y) = 0$$

$$F_x + F_z g_x(x, y) = 0$$

Because $F_z \neq 0$ near p , isolate $g_x(x, y)$:

$$g_x(x, y) = -\frac{F_x(x, y, g(x, y))}{F_z(x, y, g(x, y))} = -\frac{2x}{2z} = -\frac{x}{g(x, y)}$$

- **Differentiating with respect to y :**

Differentiate both sides of $F(x, y, g(x, y)) = 0$ with respect to y using the multivariable chain rule:

$$\frac{\partial}{\partial y} [F(x, y, g(x, y))] = \frac{\partial F}{\partial x} \frac{\partial x}{\partial y} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial y} + \frac{\partial F}{\partial z} \frac{\partial g}{\partial y} = 0$$

Since $\frac{\partial x}{\partial y} = 0$ and $\frac{\partial y}{\partial y} = 1$:

$$F_x(x, y, g(x, y)) \cdot 0 + F_y(x, y, g(x, y)) \cdot 1 + F_z(x, y, g(x, y)) g_y(x, y) = 0$$

$$F_y + F_z g_y(x, y) = 0$$

Since $F_z \neq 0$ near p , isolate $g_y(x, y)$:

$$g_y(x, y) = -\frac{F_y(x, y, g(x, y))}{F_z(x, y, g(x, y))} = -\frac{2y}{2z} = -\frac{y}{g(x, y)}$$

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6 Comprehension Check

Problem 15. Diagnose the statement

For each statement below, decide whether it is true or false.

If it is true, give a short proof or cite the appropriate result and explain why its hypotheses are satisfied. If it is false, give a counterexample and explain which implication fails.

- (a) If f is differentiable at x , then f is continuous at x .
- (b) If all partial derivatives of f exist at x , then f is differentiable at x .
- (c) If every directional derivative of f exists at x , then f is differentiable at x .
- (d) If all first partial derivatives exist in a neighborhood of x and are continuous at x , then f is differentiable at x .
- (e) A continuous image of a compact set is compact.

- (b) If all partial derivatives of f exist at x , then f is differentiable at x .

- (c) If every directional derivative of f exists at x , then f is differentiable at x .

- (d) If all first partial derivatives exist in a neighborhood of x and are continuous at x , then f is differentiable at x .

- (e) A continuous image of a compact set is compact.

- (f) A continuous image of a closed set is closed.

- (g) A closed subset of a compact set is compact.

- (h) If $Df(x)$ is invertible, then f is globally one-to-one.

- (i) If $Df(x)$ is singular, then f cannot possess a differentiable local inverse at x .

- (j) If $Df(x)$ is singular, then f cannot be locally one-to-one near x .

Problem 15: Diagnose the Statement

- (a) If f is differentiable at x , then f is continuous at x .

◦ **Verdict: True**

◦ **Proof:** By definition of differentiability, $f(x+h) - f(x) = Df(x)h + r(h)$, where $\lim_{h \rightarrow 0} \frac{\|r(h)\|}{\|h\|} = 0$. Taking norms and applying the triangle inequality:

$$\|f(x+h) - f(x)\| \leq \|Df(x)\| \|h\| + \|r(h)\|$$

As $h \rightarrow 0$, $\|Df(x)\| \|h\| \rightarrow 0$ and $\|r(h)\| = \frac{\|r(h)\|}{\|h\|} \|h\| \rightarrow 0 \cdot 0 = 0$. Thus, $\lim_{h \rightarrow 0} f(x+h) = f(x)$, proving f is continuous at x .

- (b) If all partial derivatives of f exist at x , then f is differentiable at x .

◦ **Verdict: False**

◦ **Counterexample:** Consider $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by:

$$f(x, y) = \begin{cases} 1, & \text{if } x = 0 \text{ or } y = 0 \\ 0, & \text{otherwise} \end{cases}$$

At $(0,0)$, $f_x(0,0) = \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h \rightarrow 0} \frac{1 - 0}{h} = 0$, and similarly $f_y(0,0) = 0$. Both partial derivatives exist, but f is not even continuous at $(0,0)$ along diagonal paths (e.g., along $y = x$, $f(t, t) = 0 \neq 1$), so by part (a) it cannot be differentiable.

- (c) If every directional derivative of f exists at x , then f is differentiable at x .

◦ **Verdict: False**

◦ **Counterexample:** Consider $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ from Problem 8:

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Every directional derivative $D_v f(0,0) = \frac{a^2 b}{a^2 + b^2}$ exists at the origin for $v = (a, b) \neq 0$. However, the directional derivative map $v \mapsto D_v f(0,0)$ is non-linear ($D_{(1,1)} f(0,0) = \frac{1}{2} \neq D_{(1,0)} f(0,0) + D_{(0,1)} f(0,0) = 0$), which violates the necessary linear structure $D_v f(x) = Df(x)v$ of differentiable functions.

- (d) If all first partial derivatives exist in a neighborhood of x and are continuous at x , then f is differentiable at x .

◦ **Verdict: True**

◦ **Citation / Proof:** This is the **Continuous Differentiability (C^1) Theorem** proven in Problem 9. Using the 1D Mean Value Theorem along coordinate axes increments:

$$f(x+h) - f(x) = \sum_{i=1}^n f_{x_i}(x) h_i = \sum_{i=1}^n [f_{x_i}(\xi_i) - f_{x_i}(x)] h_i$$

Dividing by $\|h\|$ and using the continuity of each f_{x_i} at x ensures that $\lim_{h \rightarrow 0} \frac{\|r(h)\|}{\|h\|} = 0$, establishing total differentiability.

- (e) A continuous image of a compact set is compact.

◦ **Verdict: True**

◦ **Citation / Proof:** This is the general topological result proven in Problem 5(a). Let K be compact and f be continuous. For any sequence $(y_k) \subseteq f(K)$, choose $x_k \in K$ with $f(x_k) = y_k$. By sequential compactness of K , there is a subsequence $x_{k_j} \rightarrow x^* \in K$. By sequential continuity of f , $y_{k_j} = f(x_{k_j}) \rightarrow f(x^*) \in f(K)$, showing that every sequence in $f(K)$ has a subsequence converging to a point in $f(K)$.

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Problem 15 (Continued)

- (f) A continuous image of a closed set is closed.

◦ **Verdict: False**

◦ **Counterexample:** Let $F = [1, \infty) \subset \mathbb{R}$, which is a closed set. Define the continuous function $f: [1, \infty) \rightarrow \mathbb{R}$ by $f(x) = \frac{1}{x}$. The image is $f(F) = (0, 1]$, which is not closed in $\mathbb{R}$ because it fails to contain its limit point 0.

- (g) A closed subset of a compact set is compact.

◦ **Verdict: True**

◦ **Proof:** Let $K \subseteq \mathbb{R}^n$ be compact and $C \subseteq K$ be closed. Let $(x_k)_{k=1}^\infty$ be an arbitrary sequence in C . Since $C \subseteq K$, (x_k) is also a sequence in the compact set K . By sequential compactness of K , there exists a subsequence (x_{k_j}) converging to some limit $x^* \in K$. Because C is closed, it contains all of its limit points, so $x^* \in C$. Thus, every sequence in C possesses a subsequence converging to a point in C , proving C is compact.

- (h) If $Df(x)$ is invertible, then f is globally one-to-one.

◦ **Verdict: False**

◦ **Counterexample:** Consider $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ given by $F(x, y) = (e^x \cos y, e^x \sin y)$ from Problem 13. The Jacobian determinant satisfies $\det D F(x, y) = e^{2x} > 0$ for all $(x, y) \in \mathbb{R}^2$, meaning $Df(x)$ is everywhere invertible. However, F is periodic with $F(x, y + 2\pi) = F(x, y)$, so $F(0,0) = F(0,2\pi) = (1,0)$, showing F is not globally one-to-one.

- (i) If $Df(x)$ is singular, then f cannot possess a differentiable local inverse at x .

◦ **Verdict: True**

◦ **Proof:** Suppose for contradiction that f has a local inverse $g = f^{-1}$ defined near $y = f(x)$ that is differentiable at y . Then $g(f(x)) = x$. Differentiating both sides via the multivariate chain rule gives:

$$Dg(f(x)) \cdot Df(x) = I_n$$

Taking determinants of both sides yields:

$$\det Dg(f(x)) \cdot \det Df(x) = \det(I_n) = 1$$

If $Df(x)$ is singular, then $\det Df(x) = 0$, leading to $0 = 1$, a contradiction. Therefore, no differentiable local inverse can exist at x .

- (j) If $Df(x)$ is singular, then f cannot be locally one-to-one near x .

◦ **Verdict: False**

◦ **Counterexample:** Consider $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^3$ at $x = 0$. The derivative is $f'(0) = 3(0)^2 = 0$ (the 1×1 Jacobian is singular). However, f is strictly monotonically increasing on all of $\mathbb{R}$, which means f is strictly injective (one-to-one) on every open neighborhood around 0,

admitting the continuous local inverse $f^{-1}(y) = y^{1/3}$.

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(k) If $U \subseteq \mathbb{R}^n$ is convex and $Df(x) = 0$ for every $x \in U$, then f is constant on U .

Problem 15 (Continued)

• (k) If $U \subseteq \mathbb{R}^n$ is convex and $Df(x) = 0$ for every $x \in U$, then f is constant on U .

◦ **Verdict: True**

◦ **Proof:** Let $a, b \in U$ be arbitrary. Because U is convex, the line segment $[a, b] = \{a + t(b - a) : t \in [0, 1]\}$ is contained entirely in U . By the Mean Value Theorem on line segments (Problem 10(b)), there exists a point $c \in (a, b) \subseteq U$ such that

$$f(b) - f(a) = Df(c)(b - a)$$

Since $Df(x) = 0$ for all $x \in U$, we have $Df(c) = 0$. Therefore, $f(b) - f(a) = 0(b - a) = 0 \Rightarrow f(b) = f(a)$. Since a and b were arbitrary, f is constant on U .

• (l) If $U \subseteq \mathbb{R}^n$ is convex and $H_f(x) = 0$ for every $x \in U$, then f is affine on U .

◦ **Verdict: True** (assuming $f \in C^2(U)$ with U open)

◦ **Proof:** The Jacobian matrix of the gradient map $\nabla f: U \rightarrow \mathbb{R}^n$ is the Hessian matrix: $D(\nabla f)(x) = H_f(x)$. Since $H_f(x) = 0$ everywhere on the convex set U , applying statement (k) component-wise to ∇f implies that $\nabla f(x)$ is constant on U , so $\nabla f(x) = a$ for some fixed $a \in \mathbb{R}^n$.

Now define the auxiliary function $g(x) = f(x) - a^T x$. Its derivative is $Dg(x) = Df(x) - a^T = \nabla f(x)^T - a^T = 0$ for all $x \in U$. By statement (k), $g(x)$ is constant on U , so $g(x) = b$ for some scalar $b \in \mathbb{R}$. Thus, $f(x) = a^T x + b$, which is affine on U .

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(l) If $U \subseteq \mathbb{R}^n$ is convex and $H_f(x) = 0$ for every $x \in U$, then f is affine on U .

Optional Extensions

The following problems are not required. They provide additional proof practice and connect the topology developed above with later results in multivariate calculus.

Extension 1. Nested compact sets

Suppose

$$K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$$

is a decreasing sequence of nonempty compact subsets of $\mathbb{R}^n$.

Prove that

$$\bigcap_{k=1}^{\infty} K_k \neq \emptyset.$$

Guidance. Choose

$$x_k \in K_k.$$

All of these points belong to K_1 , so extract a convergent subsequence.

If

$$x_{k_j} \rightarrow x,$$

fix an arbitrary N . For sufficiently large j ,

$$k_j \geq N.$$

Use the nestedness of the sets and closedness of K_N .

Extension 1: Nested Compact Sets (Cantor's Intersection Theorem)

Proposition: Let $K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$ be a decreasing sequence of non-empty compact subsets of $\mathbb{R}^n$. Then:

$$\bigcap_{k=1}^{\infty} K_k \neq \emptyset$$

Proof via Guided Subsequential Extraction:

1. Construct a Sequence:

Since each set K_k is non-empty, use the axiom of choice to select an element $x_k \in K_k$ for each $k \in \mathbb{N} = \{1, 2, 3, \dots\}$. This defines a sequence $(x_k)_{k=1}^{\infty}$.

2. Containment in the Ambient Set K_1 :

Because the sequence of sets is nested ($K_1 \supseteq K_2 \supseteq K_3 \supseteq \cdots$), we have $K_k \subseteq K_1$ for all $k \geq 1$. Consequently, every term of the sequence satisfies:

$$x_k \in K_1 \quad \forall k \in \mathbb{N}$$

3. Extract a Convergent Subsequence:

The set $K_1 \subseteq \mathbb{R}^n$ is compact, hence sequentially compact. Therefore, there exists a subsequence $(x_{k_j})_{j=1}^{\infty}$ and a limit point $x \in K_1$ such that:

$$\lim_{j \rightarrow \infty} x_{k_j} = x$$

4. Show $x \in K_N$ for an Arbitrary N :

Fix an arbitrary index $N \in \mathbb{N}$.

◦ Since $k_1 < k_2 < k_3 < \cdots$ is a strictly increasing sequence of positive integers, $k_j \rightarrow \infty$ as $j \rightarrow \infty$. Thus, there exists an integer $J \in \mathbb{N}$ such that for all $j \geq J$:

$$k_j \geq N$$

◦ By nestedness, $K_{k_j} \subseteq K_N$ whenever $k_j \geq N$. Since $x_{k_j} \in K_{k_j}$, it follows that:

$$x_{k_j} \in K_N \quad \forall j \geq J$$

◦ The tail of the subsequence $(x_{k_j})_{j=J}^{\infty}$ lies entirely within K_N . Because K_N is compact in $\mathbb{R}^n$, it is closed. A closed set contains all of its sequential limit points:

$$x = \lim_{j \rightarrow \infty} x_{k_j} \in K_N$$

5. Conclusion:

Because the choice of $N \in \mathbb{N}$ was arbitrary, $x \in K_N$ holds for every $N \geq 1$. Therefore:

$$x \in \bigcap_{N=1}^{\infty} K_N \Rightarrow \bigcap_{k=1}^{\infty} K_k \neq \emptyset \quad \blacksquare$$

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Extension 2. Invertible matrices form an open set

Identify the space of $n \times n$ real matrices with $\mathbb{R}^{n^2}$, and define

$$GL_n(\mathbb{R}) = \{A \in \mathbb{R}^{n \times n} : \det A \neq 0\}.$$

(a) Prove that $GL_n(\mathbb{R})$ is open.

(b) Suppose $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is C^1 and $Df(x^*)$ is invertible. Prove that there exists $\varepsilon > 0$ such that $Df(x)$ remains invertible whenever

$$\|x - x^*\| < \varepsilon.$$

Guidance. Regard

Extension 2: Invertible Matrices Form an Open Set

Part (a): Prove that $GL_n(\mathbb{R})$ is open

Proof:

1. Continuity of the Determinant:

Identify $\mathbb{R}^{n \times n} \cong \mathbb{R}^{n^2}$. The determinant map:

$$\det: \mathbb{R}^{n \times n} \rightarrow \mathbb{R}$$

is given by the Leibniz formula:

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) \prod_{i=1}^n A_{i,\sigma(i)}$$

Because $\det(A)$ is a multilinear polynomial in the n^2 matrix entries A_{ij} , it is a polynomial function from $\mathbb{R}^{n^2}$ to $\mathbb{R}$, and hence is **continuous**.

2. Preimage of an Open Set:

By definition:

(b) Suppose $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is C^1 and $Df(x^*)$ is invertible. Prove that there exists $\varepsilon > 0$ such that $Df(x)$ remains invertible whenever

$$\|x - x^*\| < \varepsilon.$$

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$$\det: \mathbb{R}^{n^2} \rightarrow \mathbb{R}$$

as a continuous function and write

$$GL_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\}).$$

Because $\det(A)$ is a multilinear polynomial in the n^2 matrix entries A_{ij} , it is a polynomial function from $\mathbb{R}^{n^2}$ to $\mathbb{R}$, and hence is **continuous**.

2. Preimage of an Open Set:

By definition:

$$GL_n(\mathbb{R}) = \{A \in \mathbb{R}^{n \times n} : \det A \neq 0\} = \det^{-1}(\mathbb{R} \setminus \{0\})$$

The target set $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$ is open in $\mathbb{R}$.

Because the preimage of an open set under a continuous map is open, $GL_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$ is an open subset of $\mathbb{R}^{n \times n} \cong \mathbb{R}^{n^2}$. ■

Part (b): Open Neighborhood Preservation of Invertibility for $Df(x)$

Proof.

1. Continuity of the Derivative Map:

Because $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is C^1 , each partial derivative $\frac{\partial f_i}{\partial x_j}(x)$ is continuous on $\mathbb{R}^n$. Consequently, the total derivative operator:

$$Df: \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}, \quad x \mapsto Df(x)$$

is a continuous mapping from $\mathbb{R}^n$ into the matrix space $\mathbb{R}^{n^2}$.

2. Preimage of $GL_n(\mathbb{R})$:

From Part (a), $GL_n(\mathbb{R})$ is an open subset of $\mathbb{R}^{n \times n}$. By continuity of the map Df , the preimage:

$$U = (Df)^{-1}(GL_n(\mathbb{R})) = \{x \in \mathbb{R}^n : Df(x) \in GL_n(\mathbb{R})\}$$

is an **open set** in $\mathbb{R}^n$.

3. Existence of the ε -Ball:

By hypothesis, $Df(x^*)$ is invertible, which means $Df(x^*) \in GL_n(\mathbb{R})$. Therefore, $x^* \in U$.

Because U is an open subset of the metric space $(\mathbb{R}^n, \|\cdot\|)$, every point in U is an interior point. Thus, there exists a radius $\varepsilon > 0$ such that the open ball $B(x^*, \varepsilon) \subseteq U$:

$$B(x^*, \varepsilon) = \{x \in \mathbb{R}^n : \|x - x^*\| < \varepsilon\} \subseteq U$$

Hence, for every $x \in \mathbb{R}^n$ satisfying $\|x - x^*\| < \varepsilon$, we have $x \in U \Rightarrow Df(x) \in GL_n(\mathbb{R})$, so $Df(x)$ remains invertible. ■

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